

ULTIMATE CYBERSECURITY MASTER BLUEPRINT 2026-2027

Volume 4 — Daily Curriculum: Months 9-12 + Final Consolidation

141 daily cards covering cloud, specialization, capstone, final exams and Week 49-52 integration

Edition: Volume 4
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System: 365 days, 12 academic months, 52 weeks + 1 graduation day
Profile: BS CS student in Pakistan; free/high-value resources; safe authorized labs

Academic-style self-study manual. No employment, salary, visa, certification or top-1% guarantee.

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Month 09 — Cloud, containers, Kubernetes and infrastructure security

Days D225-D252. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Cloud service models and shared responsibility D2: Cloud account structure and IAM foundations D3: Cloud networking and segmentation D4: Cloud storage, encryption and data lifecycle D5: Cloud logging, monitoring and incident response D6: Free-tier safety and cloud cost controls D7: Weekly cloud foundations review and paper design
Week 2	D1: Docker architecture and local setup D2: Dockerfile security and image hygiene D3: Container networking, volumes and secrets D4: Image scanning, SBOM and vulnerability triage D5: Container runtime security and isolation D6: Weekly container security review D7: Kubernetes architecture and control plane
Week 3	D1: Kubernetes RBAC and service accounts D2: Network policies and workload isolation D3: Pod security, secrets and admission D4: Kubernetes supply chain and audit logging D5: Local Kubernetes security lab and review D6: CI/CD security and GitHub Actions hardening D7: Infrastructure as Code security
Week 4	D1: Dependency scanning, provenance and release gates D2: Cloud configuration security review project D3: Docker image and container security review project D4: Month 9 practical examination and milestone gate D5: Monthly review and gap repairs D6: Multi-cloud comparison and vendor lock-in controls D7: Spaced revision, recovery and Month 10 preview

> **Gate:** score the Month 9 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 09 · Week 1 · daily cards

D225 · M09W1D1 · Cloud service models and shared responsibility

Phase: Academic month · **Optional date:** 2027-06-08 · **Domain resources:** CL03 / CL05

1. Day / sequence	Global Day 225 of 365; calendar-free M09W1D1; optional date 2027-06-08.
2. Exact topic	Cloud service models and shared responsibility
3. Exact subtopics	IaaS, PaaS, SaaS, physical/infrastructure/platform/application/data layers, customer/provider duties and control planes
4. Learning objective	Explain which security tasks remain the customer's responsibility in each service model
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar, Security Foundations and shared responsibility sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Azure security documentation `https://learn.microsoft.com/en-us/azure/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a responsibility matrix for a VM, managed database and serverless function; identify three common misunderstanding risks. **2.** Standard (~2 h): Build a responsibility matrix for a VM, managed database and serverless function; identify three common misunderstanding risks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a responsibility matrix for a VM, managed database and serverless function; identify three common misunderstanding risks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	shared-responsibility.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a managed service not remove the need for identity and data classification? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every row correctly separates provider and customer duties Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar, Security Foundations and shared responsibility sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D226 · M09W1D2 · Cloud account structure and IAM foundations

Phase: Academic month · **Optional date:** 2027-06-09 · **Domain resources:** CL02 / CL01

1. Day / sequence	Global Day 226 of 365; calendar-free M09W1D2; optional date 2027-06-09.
2. Exact topic	Cloud account structure and IAM foundations
3. Exact subtopics	Accounts/subscriptions, organizations, root user, principals, policies, roles, groups, temporary credentials and least privilege
4. Learning objective	Design an IAM policy from deny-by-default principles
5. Primary lecture / reading + direct URL	AWS security documentation `https://docs.aws.amazon.com/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	AWS security documentation, IAM best practices and identity sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Skill Builder `https://skillbuilder.aws/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write three synthetic IAM policies: read-only analyst, deployment role and break-glass admin; explain why one is too broad. **2.** Standard (~2 h): Write three synthetic IAM policies: read-only analyst, deployment role and break-glass admin; explain why one is too broad. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write three synthetic IAM policies: read-only analyst, deployment role and break-glass admin; explain why one is too broad. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iam-policies.json and review notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should daily work avoid long-lived root or administrator credentials? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Policies are syntactically plausible and least-privilege analysis is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS security documentation, IAM best practices and identity sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Well-Architected Security Pillar; write a five-line correction; retest next morning with a new 10-minute problem.

D227 · M09W1D3 · Cloud networking and segmentation

Phase: Academic month · **Optional date:** 2027-06-10 · **Domain resources:** CL03 / CL05

1. Day / sequence	Global Day 227 of 365; calendar-free M09W1D3; optional date 2027-06-10.
2. Exact topic	Cloud networking and segmentation
3. Exact subtopics	VPC/VNet, subnets, route tables, security groups, NACLs, private endpoints, load balancers and egress control
4. Learning objective	Draw a segmented cloud network with explicit allow/deny paths
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar, Infrastructure Protection section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Azure security documentation `https://learn.microsoft.com/en-us/azure/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a paper architecture with public/private subnets, one database and one admin path; write a rule table and failure-domain analysis. **2.** Standard (~2 h): Create a paper architecture with public/private subnets, one database and one admin path; write a rule table and failure-domain analysis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a paper architecture with public/private subnets, one database and one admin path; write a rule table and failure-domain analysis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-network.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which rule unintentionally allows broad lateral movement? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Diagram and rule table agree and all high-risk paths have compensating controls Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar, Infrastructure Protection section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D228 · M09W1D4 · Cloud storage, encryption and data lifecycle

Phase: Academic month · **Optional date:** 2027-06-11 · **Domain resources:** CL02 / C25

1. Day / sequence	Global Day 228 of 365; calendar-free M09W1D4; optional date 2027-06-11.
2. Exact topic	Cloud storage, encryption and data lifecycle
3. Exact subtopics	Object storage, block storage, buckets, ACLs/policies, public access blocks, encryption at rest/in transit, KMS keys, versioning, logging and retention
4. Learning objective	Review a synthetic storage configuration and fix public-data exposure
5. Primary lecture / reading + direct URL	AWS security documentation `https://docs.aws.amazon.com/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	AWS security documentation, data protection and encryption sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Privacy Framework `https://www.nist.gov/privacy-framework`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create ten synthetic storage resources with five misconfigurations; classify data and write remediation with verification commands. **2.** Standard (~2 h): Create ten synthetic storage resources with five misconfigurations; classify data and write remediation with verification commands. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create ten synthetic storage resources with five misconfigurations; classify data and write remediation with verification commands. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-storage-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can encryption at rest fail to protect an object that is publicly readable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All five findings are fixed in the design and encryption versus access control is distinguished Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS security documentation, data protection and encryption sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Well-Architected Security Pillar; write a five-line correction; retest next morning with a new 10-minute problem.

D229 · M09W1D5 · Cloud logging, monitoring and incident response

Phase: Academic month · **Optional date:** 2027-06-12 · **Domain resources:** B01 / CL02

1. Day / sequence	Global Day 229 of 365; calendar-free M09W1D5; optional date 2027-06-12.
2. Exact topic	Cloud logging, monitoring and incident response
3. Exact subtopics	Control-plane logs, data-plane logs, flow logs, audit trails, metric alerts, retention, tamper resistance and incident evidence
4. Learning objective	Design a minimum viable cloud detection pipeline without enabling costly services
5. Primary lecture / reading + direct URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/` **Type:** Official documentation · **Access:** Docs free; Azure/Sentinel usage may incur costs
6. Exact section / lesson locator	Microsoft Sentinel documentation for comparative cloud SIEM concepts; CL03 AWS Security Pillar Detection section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS security documentation `https://docs.aws.amazon.com/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a log-source matrix and two detection hypotheses for suspicious IAM changes and public bucket creation. **2.** Standard (~2 h): Create a log-source matrix and two detection hypotheses for suspicious IAM changes and public bucket creation. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a log-source matrix and two detection hypotheses for suspicious IAM changes and public bucket creation. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-logging-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which log is necessary to prove who changed a security policy? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Matrix includes source, owner, retention, integrity and required fields Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Sentinel documentation for comparative cloud SIEM concepts; CL03 AWS Security Pillar Detection section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kusto Query Language overview; write a five-line correction; retest next morning with a new 10-minute problem.

D230 · M09W1D6 · Free-tier safety and cloud cost controls

Phase: Academic month · **Optional date:** 2027-06-13 · **Domain resources:** CL04 / CL01

1. Day / sequence	Global Day 230 of 365; calendar-free M09W1D6; optional date 2027-06-13.
2. Exact topic	Free-tier safety and cloud cost controls
3. Exact subtopics	Free Tier types, budgets, billing alarms, account aliases, MFA, resource tags, cleanup, region selection and IaC teardown
4. Learning objective	Prepare a zero-surprise cloud lab policy before creating any resources
5. Primary lecture / reading + direct URL	AWS Free Tier `https://aws.amazon.com/free/` **Type:** Official program page · **Access:** Program terms; some services free temporarily, others always free
6. Exact section / lesson locator	AWS Free Tier official page and CL03 Well-Architected Security Pillar operational practices. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Skill Builder `https://skillbuilder.aws/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a pre-flight checklist including MFA, budget alert, resource tagging, daily cleanup and a do-not-deploy list; do not create billable resources yet. **2.** Standard (~2 h): Write a pre-flight checklist including MFA, budget alert, resource tagging, daily cleanup and a do-not-deploy list; do not create billable resources yet. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a pre-flight checklist including MFA, budget alert, resource tagging, daily cleanup and a do-not-deploy list; do not create billable resources yet. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-preflight.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which free-tier offer expires and which is always free, and why does that matter? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Checklist is complete and no cloud account has active paid resources Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Free Tier official page and CL03 Well-Architected Security Pillar operational practices"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Skill Builder; write a five-line correction; retest next morning with a new 10-minute problem.

D231 · M09W1D7 · Weekly cloud foundations review and paper design

Phase: Academic month · **Optional date:** 2027-06-14 · **Domain resources:** CL03 / CL16 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 231 of 365; calendar-free M09W1D7; optional date 2027-06-14.
2. Exact topic	Weekly cloud foundations review and paper design
3. Exact subtopics	Shared responsibility, IAM, network, storage, logging and cost controls; active recall
4. Learning objective	Demonstrate Month 9 Week 1 competence without paid infrastructure
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar review questions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Cloud Security Alliance Cloud Controls Matrix `https://cloudsecurityalliance.org/research/cloud-controls-matrix`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute architecture review of a fictional small cloud app; identify ten risks, map to CCM/CSF and propose prioritized fixes — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute architecture review of a fictional small cloud app; identify ten risks, map to CCM/CSF and propose prioritized fixes — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete a 60-minute architecture review of a fictional small cloud app; identify ten risks, map to CCM/CSF and propose prioritized fixes — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9W1-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which risk is caused by identity design rather than network design? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all recommendations are verifiable Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar review questions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 09 · Week 2 · daily cards

D232 · M09W2D1 · Docker architecture and local setup

Phase: Academic month · **Optional date:** 2027-06-15 · **Domain resources:** CL06 / CL07

1. Day / sequence	Global Day 232 of 365; calendar-free M09W2D1; optional date 2027-06-15.
2. Exact topic	Docker architecture and local setup
3. Exact subtopics	Engine, daemon, client, images, layers, containers, registries, Dockerfile, Compose, namespaces, cgroups and root risk
4. Learning objective	Build and inspect a tiny local image without exposing the daemon publicly
5. Primary lecture / reading + direct URL	Docker documentation `https://docs.docker.com/` **Type:** Official documentation · **Access:** Free; Docker Desktop licensing may apply in some organizations
6. Exact section / lesson locator	Docker documentation, Get Started and Docker Engine overview sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Docker Engine security documentation `https://docs.docker.com/engine/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Install Docker Engine or Podman in an isolated Linux VM; build a hello-world image, inspect layers and list daemon attack-surface concerns. **2.** Standard (~2 h): Install Docker Engine or Podman in an isolated Linux VM; build a hello-world image, inspect layers and list daemon attack-surface concerns. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Install Docker Engine or Podman in an isolated Linux VM; build a hello-world image, inspect layers and list daemon attack-surface concerns. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	docker-basics.md and image metadata; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is access to the Docker daemon effectively root-level access on many hosts? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Image builds and runs locally and answer explains the daemon trust boundary Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker documentation, Get Started and Docker Engine overview sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker Engine security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D233 · M09W2D2 · Dockerfile security and image hygiene

Phase: Academic month · **Optional date:** 2027-06-16 · **Domain resources:** CL07 / W13

1. Day / sequence	Global Day 233 of 365; calendar-free M09W2D2; optional date 2027-06-16.
2. Exact topic	Dockerfile security and image hygiene
3. Exact subtopics	Minimal base images, pinned versions, non-root user, package cleanup, multi-stage builds, COPY versus ADD, secrets, labels and reproducibility
4. Learning objective	Harden a deliberately poor Dockerfile and compare before/after
5. Primary lecture / reading + direct URL	Docker Engine security documentation `https://docs.docker.com/engine/security/` **Type:** Official security docs · **Access:** Free
6. Exact section / lesson locator	Docker Engine security documentation and CL12 Trivy getting-started documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rewrite a sample Dockerfile with five security improvements, build it, and scan both versions with Trivy in offline/local mode if available. **2.** Standard (~2 h): Rewrite a sample Dockerfile with five security improvements, build it, and scan both versions with Trivy in offline/local mode if available. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rewrite a sample Dockerfile with five security improvements, build it, and scan both versions with Trivy in offline/local mode if available. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	Dockerfile.before/after and scan report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which flaw remains even after vulnerability patching and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hardened image runs as non-root and has no secret or mutable build tag Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker Engine security documentation and CL12 Trivy getting-started documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D234 · M09W2D3 · Container networking, volumes and secrets

Phase: Academic month · **Optional date:** 2027-06-17 · **Domain resources:** CL06 / CL07

1. Day / sequence	Global Day 234 of 365; calendar-free M09W2D3; optional date 2027-06-17.
2. Exact topic	Container networking, volumes and secrets
3. Exact subtopics	Bridge/host/none networks, port publishing, inter-container communication, volumes, bind mounts, read-only root filesystem and secret injection
4. Learning objective	Explain how data and secrets can leak through container configuration
5. Primary lecture / reading + direct URL	Docker documentation `https://docs.docker.com/` **Type:** Official documentation · **Access:** Free; Docker Desktop licensing may apply in some organizations
6. Exact section / lesson locator	Docker documentation, networking, volumes and Compose sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Docker Engine security documentation `https://docs.docker.com/engine/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create two local containers on an isolated network, intentionally misconfigure one and then fix it; demonstrate secrets not present in image history. **2.** Standard (~2 h): Create two local containers on an isolated network, intentionally misconfigure one and then fix it; demonstrate secrets not present in image history. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create two local containers on an isolated network, intentionally misconfigure one and then fix it; demonstrate secrets not present in image history. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	container-network.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is an environment variable sometimes a weak secret-delivery mechanism? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Final configuration is isolated and image history contains no secret Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker documentation, networking, volumes and Compose sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker Engine security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D235 · M09W2D4 · Image scanning, SBOM and vulnerability triage

Phase: Academic month · **Optional date:** 2027-06-18 · **Domain resources:** W12 / E08

1. Day / sequence	Global Day 235 of 365; calendar-free M09W2D4; optional date 2027-06-18.
2. Exact topic	Image scanning, SBOM and vulnerability triage
3. Exact subtopics	CVEs, OS/package layers, SBOM, scanner limitations, false positives, base-image provenance and patch SLAs
4. Learning objective	Triage a container scan and decide what to patch, accept or investigate
5. Primary lecture / reading + direct URL	OWASP Dependency-Check `https://owasp.org/www-project-dependency-check/` **Type:** Open-source SCA tool · **Access:** Free/open-source
6. Exact section / lesson locator	OWASP Dependency-Check and CL12 Trivy documentation, scanning and output interpretation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NVD `https://nvd.nist.gov/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Scan two local images, produce SBOM-like inventory, triage three findings by reachability and context, and write remediation actions. **2.** Standard (~2 h): Scan two local images, produce SBOM-like inventory, triage three findings by reachability and context, and write remediation actions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Scan two local images, produce SBOM-like inventory, triage three findings by reachability and context, and write remediation actions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	image-scan-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a vulnerable package in an unused code path still require a documented decision? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every finding has context, owner, action and retest plan Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Dependency-Check and CL12 Trivy documentation, scanning and output interpretation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OpenSSF Scorecard; write a five-line correction; retest next morning with a new 10-minute problem.

D236 · M09W2D5 · Container runtime security and isolation

Phase: Academic month · **Optional date:** 2027-06-19 · **Domain resources:** CL07 / L04

1. Day / sequence	Global Day 236 of 365; calendar-free M09W2D5; optional date 2027-06-19.
2. Exact topic	Container runtime security and isolation
3. Exact subtopics	Capabilities, seccomp, AppArmor/SELinux, rootless mode, privileged containers, resource limits, no-new-privileges and read-only filesystem
4. Learning objective	Run a container with reduced privileges and explain each control's failure mode
5. Primary lecture / reading + direct URL	Docker Engine security documentation `https://docs.docker.com/engine/security/` **Type:** Official security docs · **Access:** Free
6. Exact section / lesson locator	Docker Engine security documentation, namespaces, capabilities, seccomp and rootless sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux man-pages project `https://man7.org/linux/man-pages/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Compare default and hardened docker run configurations; test one blocked operation and one allowed operation. **2.** Standard (~2 h): Compare default and hardened docker run configurations; test one blocked operation and one allowed operation. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Compare default and hardened docker run configurations; test one blocked operation and one allowed operation. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	runtime-hardening.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is privileged mode especially dangerous even in a lab? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hardened container works and privileged option is not used Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker Engine security documentation, namespaces, capabilities, seccomp and rootless sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D237 · M09W2D6 · Weekly container security review

Phase: Academic month · **Optional date:** 2027-06-20 · **Domain resources:** CL12 / W15

1. Day / sequence	Global Day 237 of 365; calendar-free M09W2D6; optional date 2027-06-20.
2. Exact topic	Weekly container security review
3. Exact subtopics	Docker architecture, Dockerfile, networking, volumes, secrets, scanning and runtime isolation; active recall
4. Learning objective	Demonstrate Month 9 Week 2 competence with a local image review
5. Primary lecture / reading + direct URL	Trivy documentation `https://trivy.dev/docs/` **Type:** Official scanner docs · **Access:** Free/open-source
6. Exact section / lesson locator	Trivy documentation and CL10 OWASP Kubernetes Top 10 for image/supply-chain context. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OpenSSF Scorecard `https://securityscorecards.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review an unseen Dockerfile and runtime configuration, find ten issues and implement five fixes with tests. **2.** Standard (~2 h): Review an unseen Dockerfile and runtime configuration, find ten issues and implement five fixes with tests. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review an unseen Dockerfile and runtime configuration, find ten issues and implement five fixes with tests. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9W2-practical.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding would allow host compromise rather than just container compromise? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and all fixes are tested Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Trivy documentation and CL10 OWASP Kubernetes Top 10 for image/supply-chain context"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Dependency-Check; write a five-line correction; retest next morning with a new 10-minute problem.

D238 · M09W2D7 · Kubernetes architecture and control plane

Phase: Academic month · **Optional date:** 2027-06-21 · **Domain resources:** CL08 / CL09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 238 of 365; calendar-free M09W2D7; optional date 2027-06-21.
2. Exact topic	Kubernetes architecture and control plane
3. Exact subtopics	API server, etcd, scheduler, controller manager, kubelet, kube-proxy, pods, nodes, namespaces and reconciliation loop
4. Learning objective	Explain how a desired-state request becomes a running workload
5. Primary lecture / reading + direct URL	Kubernetes official documentation `https://kubernetes.io/docs/home/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes official documentation, Concepts, Cluster Architecture and Workloads sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Install kind/minikube only if hardware permits; otherwise use the official interactive tutorial or draw the architecture and annotate trust boundaries — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Install kind/minikube only if hardware permits; otherwise use the official interactive tutorial or draw the architecture and annotate trust boundaries — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Install kind/minikube only if hardware permits; otherwise use the official interactive tutorial or draw the architecture and annotate trust boundaries — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-architecture.svg or tutorial evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which component stores cluster state and why must access be tightly controlled? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Architecture is complete and control-plane/data-plane boundaries are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes official documentation, Concepts, Cluster Architecture and Workloads sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 09 · Week 3 · daily cards

D239 · M09W3D1 · Kubernetes RBAC and service accounts

Phase: Academic month · **Optional date:** 2027-06-22 · **Domain resources:** CL09 / CL08

1. Day / sequence	Global Day 239 of 365; calendar-free M09W3D1; optional date 2027-06-22.
2. Exact topic	Kubernetes RBAC and service accounts
3. Exact subtopics	Subjects, roles, bindings, ClusterRole, ServiceAccount, verbs, resources, default permissions and privilege escalation
4. Learning objective	Write least-privilege RBAC and test denial explicitly
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation, RBAC and service accounts sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes official documentation `https://kubernetes.io/docs/home/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a namespace, service account, role and binding that can only list pods; test one allowed and three denied actions. **2.** Standard (~2 h): Create a namespace, service account, role and binding that can only list pods; test one allowed and three denied actions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a namespace, service account, role and binding that can only list pods; test one allowed and three denied actions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	rbac-manifests/ and kubectl output; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why can wildcard verbs and resources silently violate least privilege? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests show expected allow/deny and wildcards are avoided Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation, RBAC and service accounts sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D240 · M09W3D2 · Network policies and workload isolation

Phase: Academic month · **Optional date:** 2027-06-23 · **Domain resources:** CL09 / CL08

1. Day / sequence	Global Day 240 of 365; calendar-free M09W3D2; optional date 2027-06-23.
2. Exact topic	Network policies and workload isolation
3. Exact subtopics	Default allow/deny, pod selectors, namespace selectors, ingress/egress and DNS/egress exceptions
4. Learning objective	Design a policy that blocks lateral movement without breaking DNS
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation, Network Policies and Pod Security sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes official documentation `https://kubernetes.io/docs/home/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Apply a default-deny ingress policy in a local cluster, add a minimal allow rule, and verify connectivity with two test pods. **2.** Standard (~2 h): Apply a default-deny ingress policy in a local cluster, add a minimal allow rule, and verify connectivity with two test pods. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Apply a default-deny ingress policy in a local cluster, add a minimal allow rule, and verify connectivity with two test pods. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	networkpolicy.yaml and test results; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why does a default-deny ingress policy not automatically block egress? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Tests demonstrate allow and deny paths and DNS behaviour is explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation, Network Policies and Pod Security sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D241 · M09W3D3 · Pod security, secrets and admission

Phase: Academic month · **Optional date:** 2027-06-24 · **Domain resources:** CL10 / CL08

1. Day / sequence	Global Day 241 of 365; calendar-free M09W3D3; optional date 2027-06-24.
2. Exact topic	Pod security, secrets and admission
3. Exact subtopics	Security contexts, runAsNonRoot, allowPrivilegeEscalation, capabilities, seccomp, Pod Security Admission, Secrets and encryption at rest
4. Learning objective	Harden a pod manifest and explain which controls are enforced at admission
5. Primary lecture / reading + direct URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/` **Type:** Risk reference · **Access:** Free
6. Exact section / lesson locator	OWASP Kubernetes Top 10, K01 insecure workloads and K07 secrets; CL09 Kubernetes security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes official documentation `https://kubernetes.io/docs/home/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Convert a deliberately privileged pod manifest into a hardened manifest; include Secret handling caveats and test admission policy. **2.** Standard (~2 h): Convert a deliberately privileged pod manifest into a hardened manifest; include Secret handling caveats and test admission policy. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Convert a deliberately privileged pod manifest into a hardened manifest; include Secret handling caveats and test admission policy. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	pod-hardened.yaml; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a Kubernetes Secret not automatically end-to-end secure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Cluster rejects privileged settings or paper review identifies exact admission behaviour Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Kubernetes Top 10, K01 insecure workloads and K07 secrets; CL09 Kubernetes security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D242 · M09W3D4 · Kubernetes supply chain and audit logging

Phase: Academic month · **Optional date:** 2027-06-25 · **Domain resources:** CL10 / W16

1. Day / sequence	Global Day 242 of 365; calendar-free M09W3D4; optional date 2027-06-25.
2. Exact topic	Kubernetes supply chain and audit logging
3. Exact subtopics	Image provenance, registries, tags/digests, admission policy, API audit events, runtime detection and cluster upgrade responsibility
4. Learning objective	Build a cluster supply-chain trust map with evidence sources
5. Primary lecture / reading + direct URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/` **Type:** Risk reference · **Access:** Free
6. Exact section / lesson locator	OWASP Kubernetes Top 10 K02 supply chain and K09 misconfigured components. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SLSA specification `https://slsa.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw an image-to-pod trust chain, list five controls and specify which audit event would reveal a suspicious deployment. **2.** Standard (~2 h): Draw an image-to-pod trust chain, list five controls and specify which audit event would reveal a suspicious deployment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw an image-to-pod trust chain, list five controls and specify which audit event would reveal a suspicious deployment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-supply-chain.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is an image tag weaker evidence than a digest? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Map includes builder, registry, admission, runtime and audit evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Kubernetes Top 10 K02 supply chain and K09 misconfigured components"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D243 · M09W3D5 · Local Kubernetes security lab and review

Phase: Academic month · **Optional date:** 2027-06-26 · **Domain resources:** CL11 / CL10

1. Day / sequence	Global Day 243 of 365; calendar-free M09W3D5; optional date 2027-06-26.
2. Exact topic	Local Kubernetes security lab and review
3. Exact subtopics	kind/minikube setup, namespaces, RBAC, NetworkPolicy, pod security, image scanning and audit
4. Learning objective	Demonstrate Month 9 Week 3 competence with a practical cluster review
5. Primary lecture / reading + direct URL	Kubernetes Goat `https://madhuakula.com/kubernetes-goat/` **Type:** Vulnerable-by-design K8s lab · **Access:** Free/open-source
6. Exact section / lesson locator	Kubernetes Goat beginner scenarios; use only in a disposable local cluster and only if hardware permits. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete two beginner Kubernetes Goat scenarios or an equivalent manifest review; remediate each and document resource limits. **2.** Standard (~2 h): Complete two beginner Kubernetes Goat scenarios or an equivalent manifest review; remediate each and document resource limits. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete two beginner Kubernetes Goat scenarios or an equivalent manifest review; remediate each and document resource limits. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-lab-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control failed because of cluster configuration rather than manifest syntax? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and no cluster is publicly exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes Goat beginner scenarios; use only in a disposable local cluster and only if hardware permits"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Kubernetes Top 10; write a five-line correction; retest next morning with a new 10-minute problem.

D244 · M09W3D6 · CI/CD security and GitHub Actions hardening

Phase: Academic month · **Optional date:** 2027-06-27 · **Domain resources:** CL14 / W13

1. Day / sequence	Global Day 244 of 365; calendar-free M09W3D6; optional date 2027-06-27.
2. Exact topic	CI/CD security and GitHub Actions hardening
3. Exact subtopics	Pipeline trust boundaries, secrets, GITHUB_TOKEN permissions, fork pull requests, action pinning, runners, artifacts and environments
4. Learning objective	Harden a deliberately unsafe workflow and explain each change
5. Primary lecture / reading + direct URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	GitHub Actions security hardening guide, permissions, secrets and third-party actions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a safe workflow for PRJ-08 with least permissions, pinned action SHAs, tests and dependency scanning; verify no secret in logs. **2.** Standard (~2 h): Create a safe workflow for PRJ-08 with least permissions, pinned action SHAs, tests and dependency scanning; verify no secret in logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a safe workflow for PRJ-08 with least permissions, pinned action SHAs, tests and dependency scanning; verify no secret in logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	security.yml and CI run evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why should untrusted PR code not receive repository secrets? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Workflow runs and all assigned hardening controls are present Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening guide, permissions, secrets and third-party actions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D245 · M09W3D7 · Infrastructure as Code security

Phase: Academic month · **Optional date:** 2027-06-28 · **Domain resources:** CL15 / CL13 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 245 of 365; calendar-free M09W3D7; optional date 2027-06-28.
2. Exact topic	Infrastructure as Code security
3. Exact subtopics	Declarative resources, state, providers, drift, modules, secrets in state, plan/apply, Checkov policies and immutable infrastructure
4. Learning objective	Scan and remediate a synthetic IaC configuration without deploying it
5. Primary lecture / reading + direct URL	Terraform documentation `https://developer.hashicorp.com/terraform/docs` **Type:** Official documentation · **Access:** Free/open-source; HCP Terraform has paid tiers
6. Exact section / lesson locator	Terraform documentation, configuration, state and plan sections; CL13 Checkov documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Checkov documentation `https://www.checkov.io/1.Welcome/What%20is%20Checkov.html`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a synthetic Terraform file with five misconfigurations, run Checkov or an offline checklist, remediate and compare plans — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a synthetic Terraform file with five misconfigurations, run Checkov or an offline checklist, remediate and compare plans — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a synthetic Terraform file with five misconfigurations, run Checkov or an offline checklist, remediate and compare plans — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iac-review.tf and scan report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which secret could be committed if state files are mishandled? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No apply to cloud account occurs and all five issues are remediated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Terraform documentation, configuration, state and plan sections; CL13 Checkov documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Checkov documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 09 · Week 4 · daily cards

D246 · M09W4D1 · Dependency scanning, provenance and release gates

Phase: Academic month · **Optional date:** 2027-06-29 · **Domain resources:** W16 / W13

1. Day / sequence	Global Day 246 of 365; calendar-free M09W4D1; optional date 2027-06-29.
2. Exact topic	Dependency scanning, provenance and release gates
3. Exact subtopics	SBOM, lockfiles, signed builds, SLSA levels, artifacts, attestations, release approval and incident rollback
4. Learning objective	Define a release gate that blocks unverifiable artifacts
5. Primary lecture / reading + direct URL	SLSA specification `https://slsa.dev/` **Type:** Supply-chain framework · **Access:** Free
6. Exact section / lesson locator	SLSA specification, levels and provenance; W15 OpenSSF Scorecard checks. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a release checklist with artifact hashes, provenance evidence, vulnerability thresholds and rollback owner. **2.** Standard (~2 h): Create a release checklist with artifact hashes, provenance evidence, vulnerability thresholds and rollback owner. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a release checklist with artifact hashes, provenance evidence, vulnerability thresholds and rollback owner. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	release-gate.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a green unit-test job insufficient to trust a released artifact? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Checklist includes build identity, integrity, vulnerability and rollback evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SLSA specification, levels and provenance; W15 OpenSSF Scorecard checks"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D247 · M09W4D2 · Cloud configuration security review project

Phase: Academic month · **Optional date:** 2027-06-30 · **Domain resources:** CL03 / CL02

1. Day / sequence	Global Day 247 of 365; calendar-free M09W4D2; optional date 2027-06-30.
2. Exact topic	Cloud configuration security review project
3. Exact subtopics	Asset inventory, IAM, network, storage, logging, encryption, detection and cost boundaries
4. Learning objective	Produce a professional cloud configuration review using synthetic architecture/data
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar review questions and CL16 Cloud Controls Matrix selected controls. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS security documentation `https://docs.aws.amazon.com/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review a fictional ten-resource cloud architecture, identify twelve misconfigurations, prioritize and provide verification steps; do not deploy paid resources. **2.** Standard (~2 h): Review a fictional ten-resource cloud architecture, identify twelve misconfigurations, prioritize and provide verification steps; do not deploy paid resources. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review a fictional ten-resource cloud architecture, identify twelve misconfigurations, prioritize and provide verification steps; do not deploy paid resources. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-13 report and risk register; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding has the highest blast radius and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report passes the 100-point rubric and no live cloud resources were required Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar review questions and CL16 Cloud Controls Matrix selected controls"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D248 · M09W4D3 · Docker image and container security review project

Phase: Academic month · **Optional date:** 2027-07-01 · **Domain resources:** CL12 / W12

1. Day / sequence	Global Day 248 of 365; calendar-free M09W4D3; optional date 2027-07-01.
2. Exact topic	Docker image and container security review project
3. Exact subtopics	Image layers, Dockerfile, secrets, runtime privileges, networking, volumes, scanning and SBOM
4. Learning objective	Produce an image/container security review with reproducible local evidence
5. Primary lecture / reading + direct URL	Trivy documentation `https://trivy.dev/docs/` **Type:** Official scanner docs · **Access:** Free/open-source
6. Exact section / lesson locator	Trivy documentation and CL07 Docker Engine security documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Dependency-Check `https://owasp.org/www-project-dependency-check/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review one intentionally weak local image/container, record ten findings, fix five, rescan and document residual risk. **2.** Standard (~2 h): Review one intentionally weak local image/container, record ten findings, fix five, rescan and document residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review one intentionally weak local image/container, record ten findings, fix five, rescan and document residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-14 report and before/after scan; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding was a false positive or not exploitable and how did you decide? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Containers remain isolated and all fixes are verified by retest Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Trivy documentation and CL07 Docker Engine security documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Dependency-Check; write a five-line correction; retest next morning with a new 10-minute problem.

D249 · M09W4D4 · Month 9 practical examination and milestone gate

Phase: Academic month · **Optional date:** 2027-07-02 · **Domain resources:** CL03 / CL09

1. Day / sequence	Global Day 249 of 365; calendar-free M09W4D4; optional date 2027-07-02.
2. Exact topic	Month 9 practical examination and milestone gate
3. Exact subtopics	Cloud shared responsibility, IAM, storage, logging, Docker, Kubernetes, CI/CD and IaC; practical scenario
4. Learning objective	Demonstrate infrastructure-security competence under time pressure
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar and CL08 Kubernetes concepts review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical with an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; identify and fix prioritized risks. **2.** Standard (~2 h): Complete a 150-minute practical with an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; identify and fix prioritized risks. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical with an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; identify and fix prioritized risks. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which risk crosses from workload compromise to control-plane compromise? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100, milestone gate passed, practical at least 18/25 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar and CL08 Kubernetes concepts review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D250 · M09W4D5 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-07-03 · **Domain resources:** CL09 / C20

1. Day / sequence	Global Day 250 of 365; calendar-free M09W4D5; optional date 2027-07-03.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	IAM reasoning, Docker privileges, Kubernetes RBAC/NetworkPolicy, CI secrets and IaC state; active recall
4. Learning objective	Repair Month 9 gaps and choose specialization evidence for Month 10
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation and CL14 GitHub Actions hardening guide. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, update PRJ-13/14, write ten flashcards and score your interest/results in cloud security versus detection engineering. **2.** Standard (~2 h): Rework failed assessment items, update PRJ-13/14, write ten flashcards and score your interest/results in cloud security versus detection engineering. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, update PRJ-13/14, write ten flashcards and score your interest/results in cloud security versus detection engineering. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9-review.md and specialization-prep.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which infrastructure topic would be most valuable as a secondary skill? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and scorecard uses actual project evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation and CL14 GitHub Actions hardening guide"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D251 · M09W4D6 · Multi-cloud comparison and vendor lock-in controls

Phase: Academic month · **Optional date:** 2027-07-04 · **Domain resources:** CL05 / CL02

1. Day / sequence	Global Day 251 of 365; calendar-free M09W4D6; optional date 2027-07-04.
2. Exact topic	Multi-cloud comparison and vendor lock-in controls
3. Exact subtopics	AWS/Azure equivalents for identity, network, storage, logging, KMS and incident response; portable controls and service-specific gaps
4. Learning objective	Translate one AWS control design into Azure terminology without assuming identical defaults
5. Primary lecture / reading + direct URL	Microsoft Azure security documentation `https://learn.microsoft.com/en-us/azure/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Microsoft Azure security documentation, identity, network, data protection and monitoring sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS security documentation `https://docs.aws.amazon.com/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a side-by-side control table for ten AWS/Azure concepts and identify three behaviours that must be verified in current docs. **2.** Standard (~2 h): Build a side-by-side control table for ten AWS/Azure concepts and identify three behaviours that must be verified in current docs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a side-by-side control table for ten AWS/Azure concepts and identify three behaviours that must be verified in current docs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	multicloud-controls.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control name sounds equivalent but has different default behaviour? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table uses current official docs and explicitly marks unverified assumptions Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft Azure security documentation, identity, network, data protection and monitoring sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D252 · M09W4D7 · Spaced revision, recovery and Month 10 preview

Phase: Academic month · **Optional date:** 2027-07-05 · **Domain resources:** CL04 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 252 of 365; calendar-free M09W4D7; optional date 2027-07-05.
2. Exact topic	Spaced revision, recovery and Month 10 preview
3. Exact subtopics	Cloud responsibility, IAM, Docker/Kubernetes controls, CI/CD and IaC retrieval; lab cleanup
4. Learning objective	Preserve infrastructure-security knowledge and prepare specialization selection
5. Primary lecture / reading + direct URL	AWS Free Tier `https://aws.amazon.com/free/` **Type:** Official program page · **Access:** Program terms; some services free temporarily, others always free
6. Exact section / lesson locator	AWS Free Tier and CL01 AWS Skill Builder free course catalogue. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, remove all local test containers/clusters or save clean snapshots, and read the specialization comparison in Volume 7 — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, remove all local test containers/clusters or save clean snapshots, and read the specialization comparison in Volume 7 — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, remove all local test containers/clusters or save clean snapshots, and read the specialization comparison in Volume 7 — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M9-consolidation.md and cleanup log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which specialization best matches your strongest project evidence so far? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No active billable resources remain and decision notes cite at least three projects Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Free Tier and CL01 AWS Skill Builder free course catalogue"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Skill Builder; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 — Specialization selection and focused technical development

Days D253-D280. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Specialization decision framework and evidence audit D2: Ten specialization maps and prerequisite reality D3: Primary specialization recommendation: detection engineering and security automation D4: Complementary secondary skill: cloud and container security D5: Designing a four-week specialization tasting sprint D6: Baseline assessment for specialization sprint D7: Weekly decision review and rest
Week 2	D1: Detection engineering sprint: telemetry requirements D2: Detection engineering sprint: rule design and unit tests D3: Detection engineering sprint: backtesting and tuning D4: Detection engineering sprint: automation and CI D5: Detection engineering sprint: threat hunting with the rule D6: Detection engineering sprint: documentation and defence D7: Weekly detection sprint review and score
Week 3	D1: Cloud security sprint: IAM threat modelling D2: Cloud security sprint: network and storage detections D3: Cloud security sprint: Kubernetes RBAC and NetworkPolicy review D4: Cloud security sprint: IaC and CI hardening lab D5: Cloud security sprint: incident in a containerized app D6: Cloud security sprint: cost, resilience and recovery D7: Weekly cloud sprint review and score
Week 4	D1: Decision review with a technical peer or self-defence D2: Specialization project requirements and architecture D3: Specialization literature and documentation reading D4: Independent problem-solving baseline with no walkthrough D5: Month 10 practical examination and gate D6: Monthly review and gap repairs D7: Spaced revision, recovery and Month 11 preview

> **Gate:** score the Month 10 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 10 · Week 1 · daily cards

D253 · M10W1D1 · Specialization decision framework and evidence audit

Phase: Academic month · **Optional date:** 2027-07-06 · **Domain resources:** C20 / A08

1. Day / sequence	Global Day 253 of 365; calendar-free M10W1D1; optional date 2027-07-06.
2. Exact topic	Specialization decision framework and evidence audit
3. Exact subtopics	Interest, demonstrated results, technical depth, programming comfort, local opportunity, feedback and opportunity cost
4. Learning objective	Choose a primary and secondary specialization from evidence rather than trends
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST NICE Workforce Framework, work roles, tasks, knowledge and skills. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a weighted decision matrix for ten paths using scores from M1-M9 assessments, project quality and genuine interest. **2.** Standard (~2 h): Build a weighted decision matrix for ten paths using scores from M1-M9 assessments, project quality and genuine interest. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a weighted decision matrix for ten paths using scores from M1-M9 assessments, project quality and genuine interest. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-decision.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which score is based on evidence and which is based on assumption? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Matrix is transparent, weighted and identifies a primary plus a complementary secondary Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST NICE Workforce Framework, work roles, tasks, knowledge and skills"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D254 · M10W1D2 · Ten specialization maps and prerequisite reality

Phase: Academic month · **Optional date:** 2027-07-07 · **Domain resources:** B15 / C20

1. Day / sequence	Global Day 254 of 365; calendar-free M10W1D2; optional date 2027-07-07.
2. Exact topic	Ten specialization maps and prerequisite reality
3. Exact subtopics	SOC, detection engineering, DFIR, web/API security, pentesting, cloud, security engineering, GRC, vulnerability research and malware/RE
4. Learning objective	Compare daily work, tools, programming, portfolio and realistic Year 1 limits
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and A21 PEN-200 page for two contrasting role examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a one-page comparison of ten paths with required prerequisites, first job evidence and what 12 months does not prove. **2.** Standard (~2 h): Write a one-page comparison of ten paths with required prerequisites, first job evidence and what 12 months does not prove. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a one-page comparison of ten paths with required prerequisites, first job evidence and what 12 months does not prove. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-map.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which path would require years of low-level experience before credible expertise? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No path is described as mastered and prerequisites are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and A21 PEN-200 page for two contrasting role examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D255 · M10W1D3 · Primary specialization recommendation: detection engineering and security automation

Phase: Academic month · **Optional date:** 2027-07-08 · **Domain resources:** B03 / B14

1. Day / sequence	Global Day 255 of 365; calendar-free M10W1D3; optional date 2027-07-08.
2. Exact topic	Primary specialization recommendation: detection engineering and security automation
3. Exact subtopics	Detection lifecycle, telemetry, KQL/Sigma, Python automation, cloud logs, measurement and communication
4. Learning objective	Validate whether this recommended primary fits your evidence and interest
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B04 Sigma getting-started documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a personal fit memo for detection engineering with three supportive and three contradictory pieces of project evidence. **2.** Standard (~2 h): Write a personal fit memo for detection engineering with three supportive and three contradictory pieces of project evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a personal fit memo for detection engineering with three supportive and three contradictory pieces of project evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	primary-fit-memo.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which project result would convince a working detection engineer of your potential? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Recommendation is justified but treated as revisable after the Month 10 sprint Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B04 Sigma getting-started documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D256 · M10W1D4 · Complementary secondary skill: cloud and container security

Phase: Academic month · **Optional date:** 2027-07-09 · **Domain resources:** CL03 / B01

1. Day / sequence	Global Day 256 of 365; calendar-free M10W1D4; optional date 2027-07-09.
2. Exact topic	Complementary secondary skill: cloud and container security
3. Exact subtopics	Cloud IAM, Kubernetes, Docker, CI/CD, IaC, logging and detection overlap
4. Learning objective	Show how the secondary skill strengthens the primary specialization
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar and CL09 Kubernetes security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a skill-integration map connecting cloud logs, containers, detections and automation projects. **2.** Standard (~2 h): Draw a skill-integration map connecting cloud logs, containers, detections and automation projects. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a skill-integration map connecting cloud logs, containers, detections and automation projects. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	skill-integration.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which cloud telemetry source would be most useful for your detection work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Map has at least six concrete connections and no vague tool names Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar and CL09 Kubernetes security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D257 · M10W1D5 · Designing a four-week specialization tasting sprint

Phase: Academic month · **Optional date:** 2027-07-10 · **Domain resources:** C06 / F09

1. Day / sequence	Global Day 257 of 365; calendar-free M10W1D5; optional date 2027-07-10.
2. Exact topic	Designing a four-week specialization tasting sprint
3. Exact subtopics	SMART objectives, hypotheses, timeboxing, deliverables, evidence, feedback and stopping criteria
4. Learning objective	Create a fair experiment to compare two specializations without switching randomly
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0, Govern/Identify/Protect/Detect/Respond/Recover functions as outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Define two two-week sprints with measurable deliverables, rubrics, daily logs and a final decision review. **2.** Standard (~2 h): Define two two-week sprints with measurable deliverables, rubrics, daily logs and a final decision review. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Define two two-week sprints with measurable deliverables, rubrics, daily logs and a final decision review. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sprint-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence would cause you to reject your preferred path? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes falsifiable criteria and a fixed decision date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0, Govern/Identify/Protect/Detect/Respond/Recover functions as outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D258 · M10W1D6 · Baseline assessment for specialization sprint

Phase: Academic month · **Optional date:** 2027-07-11 · **Domain resources:** B04 / B02

1. Day / sequence	Global Day 258 of 365; calendar-free M10W1D6; optional date 2027-07-11.
2. Exact topic	Baseline assessment for specialization sprint
3. Exact subtopics	Detection rule quality, query fluency, cloud misconfiguration reasoning, report clarity and independent debugging
4. Learning objective	Establish a baseline so improvement can be measured
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started rule structure and CL03 Security Pillar review questions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute baseline: one detection rule with test data, one cloud config review and one written explanation; score with the standard rubric. **2.** Standard (~2 h): Complete a 90-minute baseline: one detection rule with test data, one cloud config review and one written explanation; score with the standard rubric. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute baseline: one detection rule with test data, one cloud config review and one written explanation; score with the standard rubric. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-baseline.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which task took longest and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Baseline is scored with evidence and no tutorial assistance Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started rule structure and CL03 Security Pillar review questions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D259 · M10W1D7 · Weekly decision review and rest

Phase: Academic month · **Optional date:** 2027-07-12 · **Domain resources:** C20 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 259 of 365; calendar-free M10W1D7; optional date 2027-07-12.
2. Exact topic	Weekly decision review and rest
3. Exact subtopics	Specialization map, evidence, decision framework, primary/secondary choice, sprint plan and baseline; active recall
4. Learning objective	Consolidate Month 10 Week 1 and commit to a fair experiment
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role pages and C18 ATT&CK matrix for role-to-skill mapping. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 15 retrieval questions, review baseline errors, adjust sprint difficulty and rest or take a light walkthrough of Week 2 tasks — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 15 retrieval questions, review baseline errors, adjust sprint difficulty and rest or take a light walkthrough of Week 2 tasks — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 15 retrieval questions, review baseline errors, adjust sprint difficulty and rest or take a light walkthrough of Week 2 tasks — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10W1-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which baseline weakness is most important for your chosen primary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Sprint plan is realistic and no decision changes without documented evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role pages and C18 ATT&CK matrix for role-to-skill mapping"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 · Week 2 · daily cards

D260 · M10W2D1 · Detection engineering sprint: telemetry requirements

Phase: Academic month · **Optional date:** 2027-07-13 · **Domain resources:** C18 / B05

1. Day / sequence	Global Day 260 of 365; calendar-free M10W2D1; optional date 2027-07-13.
2. Exact topic	Detection engineering sprint: telemetry requirements
3. Exact subtopics	Data source, event schema, field coverage, time synchronization, entity mapping and blind spots
4. Learning objective	Define telemetry needed for a specific adversary behaviour
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	MITRE ATT&CK technique pages and data sources. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Pick one technique, create a telemetry requirements table and identify which fields are missing from your synthetic logs. **2.** Standard (~2 h): Pick one technique, create a telemetry requirements table and identify which fields are missing from your synthetic logs. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Pick one technique, create a telemetry requirements table and identify which fields are missing from your synthetic logs. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	telemetry-requirements.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Why is a technique description insufficient for writing a detection? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Table maps each detection field to a source and reliability Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MITRE ATT&CK technique pages and data sources"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D261 · M10W2D2 · Detection engineering sprint: rule design and unit tests

Phase: Academic month · **Optional date:** 2027-07-14 · **Domain resources:** B04 / B03

1. Day / sequence	Global Day 261 of 365; calendar-free M10W2D2; optional date 2027-07-14.
2. Exact topic	Detection engineering sprint: rule design and unit tests
3. Exact subtopics	Modifiers, conditions, correlations, thresholds, exceptions, metadata and test cases
4. Learning objective	Write a detection with explicit true/false-positive tests
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started guide and B14 Elastic detection rules docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement one advanced Sigma rule for a synthetic lab behaviour, create ten positive/negative events and validate conversion to a query. **2.** Standard (~2 h): Implement one advanced Sigma rule for a synthetic lab behaviour, create ten positive/negative events and validate conversion to a query. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement one advanced Sigma rule for a synthetic lab behaviour, create ten positive/negative events and validate conversion to a query. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	advanced-rule.yml and tests; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test event represents a realistic benign lookalike? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule passes all tests and conversion output is explained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started guide and B14 Elastic detection rules docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D262 · M10W2D3 · Detection engineering sprint: backtesting and tuning

Phase: Academic month · **Optional date:** 2027-07-15 · **Domain resources:** B14 / B02

1. Day / sequence	Global Day 262 of 365; calendar-free M10W2D3; optional date 2027-07-15.
2. Exact topic	Detection engineering sprint: backtesting and tuning
3. Exact subtopics	Historical data, prevalence, benign baselines, alert volume, precision proxy, suppression and exception design
4. Learning objective	Measure rule behaviour on a larger synthetic dataset without hiding true positives
5. Primary lecture / reading + direct URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html` **Type:** Official docs / open-source rules · **Access:** Free docs/open-source rules; Elastic Stack may have paid features
6. Exact section / lesson locator	Elastic detection rules, exceptions and alert investigation docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Generate 1,000 synthetic events with known labels, run the rule, calculate precision/recall-like measures and tune with narrow exceptions. **2.** Standard (~2 h): Generate 1,000 synthetic events with known labels, run the rule, calculate precision/recall-like measures and tune with narrow exceptions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Generate 1,000 synthetic events with known labels, run the rule, calculate precision/recall-like measures and tune with narrow exceptions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	backtest-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which exception could hide a true positive variant? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Metrics are computed from labelled data and tuning is justified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Elastic detection rules, exceptions and alert investigation docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D263 · M10W2D4 · Detection engineering sprint: automation and CI

Phase: Academic month · **Optional date:** 2027-07-16 · **Domain resources:** P05 / B03

1. Day / sequence	Global Day 263 of 365; calendar-free M10W2D4; optional date 2027-07-16.
2. Exact topic	Detection engineering sprint: automation and CI
3. Exact subtopics	Detection-as-code repository, linting, schema validation, tests, versioning, review and deployment simulation
4. Learning objective	Build a repeatable workflow for rule quality
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and F09 Pro Git branching sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create CI-like local scripts that validate Sigma YAML, run test events, produce a report and fail on regression. **2.** Standard (~2 h): Create CI-like local scripts that validate Sigma YAML, run test events, produce a report and fail on regression. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create CI-like local scripts that validate Sigma YAML, run test events, produce a report and fail on regression. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	detection-ci/ repository; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test should block a rule deployment? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Workflow fails on intentionally broken rule and passes on fixed version Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and F09 Pro Git branching sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D264 · M10W2D5 · Detection engineering sprint: threat hunting with the rule

Phase: Academic month · **Optional date:** 2027-07-17 · **Domain resources:** B15 / C18

1. Day / sequence	Global Day 264 of 365; calendar-free M10W2D5; optional date 2027-07-17.
2. Exact topic	Detection engineering sprint: threat hunting with the rule
3. Exact subtopics	Hypothesis, query, pivots, temporal correlation, entity risk and negative results
4. Learning objective	Use the new rule to answer an investigative question rather than just generate an alert
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide threat hunting section and B02 KQL operators. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a synthetic hunt dataset, run three KQL-style queries, document pivots and state what remains unknown. **2.** Standard (~2 h): Create a synthetic hunt dataset, run three KQL-style queries, document pivots and state what remains unknown. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a synthetic hunt dataset, run three KQL-style queries, document pivots and state what remains unknown. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	hunt-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which data gap prevents a confident conclusion? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hunt is reproducible and includes negative-result interpretation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide threat hunting section and B02 KQL operators"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D265 · M10W2D6 · Detection engineering sprint: documentation and defence

Phase: Academic month · **Optional date:** 2027-07-18 · **Domain resources:** B03 / B14

1. Day / sequence	Global Day 265 of 365; calendar-free M10W2D6; optional date 2027-07-18.
2. Exact topic	Detection engineering sprint: documentation and defence
3. Exact subtopics	Rule rationale, ATT&CK mapping, assumptions, test evidence, known bypasses and reviewer questions
4. Learning objective	Defend the detection to a skeptical peer in writing
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ rule metadata guidance and C18 ATT&CK technique page. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a two-page detection design document, list three bypasses and answer five reviewer questions from a self-critique checklist. **2.** Standard (~2 h): Write a two-page detection design document, list three bypasses and answer five reviewer questions from a self-critique checklist. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a two-page detection design document, list three bypasses and answer five reviewer questions from a self-critique checklist. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	detection-design.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which bypass is most realistic and what additional data would detect it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Document is technically specific and includes a mitigation or accepted limitation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ rule metadata guidance and C18 ATT&CK technique page"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D266 · M10W2D7 · Weekly detection sprint review and score

Phase: Academic month · **Optional date:** 2027-07-19 · **Domain resources:** B04 / C18 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 266 of 365; calendar-free M10W2D7; optional date 2027-07-19.
2. Exact topic	Weekly detection sprint review and score
3. Exact subtopics	Telemetry, rule design, backtesting, CI, hunting and defence; independent evaluation
4. Learning objective	Measure improvement against the Week 1 baseline and compare specialization fit
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and B14 Elastic detection rule docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the week's artifacts with the 100-point rubric, repair one major gap, and write a one-page reaction on interest and difficulty — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the week's artifacts with the 100-point rubric, repair one major gap, and write a one-page reaction on interest and difficulty — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the week's artifacts with the 100-point rubric, repair one major gap, and write a one-page reaction on interest and difficulty — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10W2-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which part of detection engineering felt effortful but rewarding? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score is evidence-based and a follow-up task is scheduled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and B14 Elastic detection rule docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 · Week 3 · daily cards

D267 · M10W3D1 · Cloud security sprint: IAM threat modelling

Phase: Academic month · **Optional date:** 2027-07-20 · **Domain resources:** CL02 / C17

1. Day / sequence	Global Day 267 of 365; calendar-free M10W3D1; optional date 2027-07-20.
2. Exact topic	Cloud security sprint: IAM threat modelling
3. Exact subtopics	Principal, action, resource, condition, privilege escalation, confused deputy, cross-account access and logging
4. Learning objective	Threat-model an IAM design before writing policies
5. Primary lecture / reading + direct URL	AWS security documentation `https://docs.aws.amazon.com/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	AWS security documentation IAM best practices and CL03 Security Pillar identity section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw an IAM attack graph for a fictional app, identify five escalation paths and propose least-privilege fixes. **2.** Standard (~2 h): Draw an IAM attack graph for a fictional app, identify five escalation paths and propose least-privilege fixes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw an IAM attack graph for a fictional app, identify five escalation paths and propose least-privilege fixes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iam-threat-model.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which permission is dangerous only in combination with another? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Graph is plausible and mitigations break specific edges Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS security documentation IAM best practices and CL03 Security Pillar identity section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS Well-Architected Security Pillar; write a five-line correction; retest next morning with a new 10-minute problem.

D268 · M10W3D2 · Cloud security sprint: network and storage detections

Phase: Academic month · **Optional date:** 2027-07-21 · **Domain resources:** CL03 / B01

1. Day / sequence	Global Day 268 of 365; calendar-free M10W3D2; optional date 2027-07-21.
2. Exact topic	Cloud security sprint: network and storage detections
3. Exact subtopics	Public exposure, policy changes, unusual data access, egress anomalies and log sources
4. Learning objective	Design cloud detections that connect infrastructure misconfiguration to monitoring
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar, Detection and Data Protection sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft Sentinel documentation `https://learn.microsoft.com/en-us/azure/sentinel/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write two detection hypotheses with required cloud log fields, one query-like pseudo-rule and a validation plan. **2.** Standard (~2 h): Write two detection hypotheses with required cloud log fields, one query-like pseudo-rule and a validation plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write two detection hypotheses with required cloud log fields, one query-like pseudo-rule and a validation plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-detections.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which detection depends on a log source that is not enabled by default? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Hypotheses specify fields, owner and enabling steps Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar, Detection and Data Protection sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D269 · M10W3D3 · Cloud security sprint: Kubernetes RBAC and NetworkPolicy review

Phase: Academic month · **Optional date:** 2027-07-22 · **Domain resources:** CL09 / CL10

1. Day / sequence	Global Day 269 of 365; calendar-free M10W3D3; optional date 2027-07-22.
2. Exact topic	Cloud security sprint: Kubernetes RBAC and NetworkPolicy review
3. Exact subtopics	Role bindings, service accounts, default-deny, egress, namespace isolation and admission controls
4. Learning objective	Review a cluster design for both prevention and detection of privilege misuse
5. Primary lecture / reading + direct URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Kubernetes security documentation, RBAC, NetworkPolicy and audit sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Kubernetes Top 10 `https://owasp.org/www-project-kubernetes-top-ten/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze ten synthetic manifests, find five privilege or segmentation issues, and write one audit-event detection hypothesis. **2.** Standard (~2 h): Analyze ten synthetic manifests, find five privilege or segmentation issues, and write one audit-event detection hypothesis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze ten synthetic manifests, find five privilege or segmentation issues, and write one audit-event detection hypothesis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	k8s-skill-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which RBAC binding creates cross-namespace access? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Findings are technically correct and no live cluster is exposed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Kubernetes security documentation, RBAC, NetworkPolicy and audit sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Kubernetes official documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D270 · M10W3D4 · Cloud security sprint: IaC and CI hardening lab

Phase: Academic month · **Optional date:** 2027-07-23 · **Domain resources:** CL14 / W16

1. Day / sequence	Global Day 270 of 365; calendar-free M10W3D4; optional date 2027-07-23.
2. Exact topic	Cloud security sprint: IaC and CI hardening lab
3. Exact subtopics	Static IaC scanning, workflow permissions, action pinning, secret handling and plan review
4. Learning objective	Combine preventive and detective controls in a software delivery pipeline
5. Primary lecture / reading + direct URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	GitHub Actions security hardening and CL13 Checkov documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SLSA specification `https://slsa.dev/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a local pipeline that scans a synthetic Terraform/Kubernetes repo, fails on three issues and produces a remediation report. **2.** Standard (~2 h): Build a local pipeline that scans a synthetic Terraform/Kubernetes repo, fails on three issues and produces a remediation report. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a local pipeline that scans a synthetic Terraform/Kubernetes repo, fails on three issues and produces a remediation report. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	iac-ci-demo/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control prevents the issue and which detects recurrence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline passes after fixes and report distinguishes prevention from detection Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening and CL13 Checkov documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D271 · M10W3D5 · Cloud security sprint: incident in a containerized app

Phase: Academic month · **Optional date:** 2027-07-24 · **Domain resources:** C07 / B02

1. Day / sequence	Global Day 271 of 365; calendar-free M10W3D5; optional date 2027-07-24.
2. Exact topic	Cloud security sprint: incident in a containerized app
3. Exact subtopics	Container logs, process metadata, IAM audit events, network evidence and timeline correlation
4. Learning objective	Investigate a synthetic cloud/container incident with incomplete data
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 incident response lifecycle and CL03 Security Pillar incident response. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze a provided synthetic log bundle, build a timeline, identify the likely root cause and state evidence gaps. **2.** Standard (~2 h): Analyze a provided synthetic log bundle, build a timeline, identify the likely root cause and state evidence gaps. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze a provided synthetic log bundle, build a timeline, identify the likely root cause and state evidence gaps. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-incident-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which claim is probable but not proven by the available telemetry? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is consistent and uncertainty is explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 incident response lifecycle and CL03 Security Pillar incident response"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D272 · M10W3D6 · Cloud security sprint: cost, resilience and recovery

Phase: Academic month · **Optional date:** 2027-07-25 · **Domain resources:** C27 / CL01

1. Day / sequence	Global Day 272 of 365; calendar-free M10W3D6; optional date 2027-07-25.
2. Exact topic	Cloud security sprint: cost, resilience and recovery
3. Exact subtopics	Resource cleanup, backups, multi-AZ concepts, recovery objectives, budgets, quotas and blast-radius reduction
4. Learning objective	Add operational realism to a cloud security design
5. Primary lecture / reading + direct URL	NIST SP 800-34 Revision 1 `https://csrc.nist.gov/pubs/sp/800/34/r1/final` **Type:** Official continuity guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-34r1 and CL04 AWS Free Tier/current cost-control documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Skill Builder `https://skillbuilder.aws/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Revise the fictional architecture for resilience and cost limits; write one restore test and one budget alert plan. **2.** Standard (~2 h): Revise the fictional architecture for resilience and cost limits; write one restore test and one budget alert plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Revise the fictional architecture for resilience and cost limits; write one restore test and one budget alert plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cloud-resilience.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which resilience control adds a new security dependency? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Design includes RTO/RPO, validation and teardown evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-34r1 and CL04 AWS Free Tier/current cost-control documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D273 · M10W3D7 · Weekly cloud sprint review and score

Phase: Academic month · **Optional date:** 2027-07-26 · **Domain resources:** CL03 / B15 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 273 of 365; calendar-free M10W3D7; optional date 2027-07-26.
2. Exact topic	Weekly cloud sprint review and score
3. Exact subtopics	IAM, network/storage, Kubernetes, IaC/CI, incident response and resilience; independent evaluation
4. Learning objective	Compare cloud sprint results with detection sprint results using the same rubric
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar review questions and CL14 GitHub Actions hardening. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score all artifacts, repair one major gap and write a one-page comparison of the two sprints — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score all artifacts, repair one major gap and write a one-page comparison of the two sprints — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score all artifacts, repair one major gap and write a one-page comparison of the two sprints — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10W3-review.md and comparison.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which sprint produced stronger evidence of independent problem-solving? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Both sprints are scored with the same 100-point rubric Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar review questions and CL14 GitHub Actions hardening"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Month 10 · Week 4 · daily cards

D274 · M10W4D1 · Decision review with a technical peer or self-defence

Phase: Academic month · **Optional date:** 2027-07-27 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 274 of 365; calendar-free M10W4D1; optional date 2027-07-27.
2. Exact topic	Decision review with a technical peer or self-defence
3. Exact subtopics	Evidence, assumptions, opportunity cost, local job market constraints and 12-month limits
4. Learning objective	Make a defensible final specialization decision for Months 11-12
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Workforce Framework and official role-skill resources for selected roles. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Present your two sprint portfolios to a peer/ teacher or record a self-defence; answer five challenge questions and update the decision matrix. **2.** Standard (~2 h): Present your two sprint portfolios to a peer/teacher or record a self-defence; answer five challenge questions and update the decision matrix. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Present your two sprint portfolios to a peer/teacher or record a self-defence; answer five challenge questions and update the decision matrix. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	specialization-final-decision.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence was strongest against your preferred choice? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Final decision is documented with primary, secondary and review date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Workforce Framework and official role-skill resources for selected roles"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D275 · M10W4D2 · Specialization project requirements and architecture

Phase: Academic month · **Optional date:** 2027-07-28 · **Domain resources:** C17 / B03

1. Day / sequence	Global Day 275 of 365; calendar-free M10W4D2; optional date 2027-07-28.
2. Exact topic	Specialization project requirements and architecture
3. Exact subtopics	Problem statement, users, assets, threat model, data sources, architecture, metrics, constraints and ethics
4. Learning objective	Define a Month 11 capstone aligned with the chosen specialization
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling and C06 NIST CSF 2.0 outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a two-page capstone specification with testable objectives, security boundaries, synthetic data plan and success metrics. **2.** Standard (~2 h): Write a two-page capstone specification with testable objectives, security boundaries, synthetic data plan and success metrics. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a two-page capstone specification with testable objectives, security boundaries, synthetic data plan and success metrics. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	capstone-spec.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement is both measurable and genuinely difficult? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Specification is approved by the self-review checklist and has no illegal targets Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling and C06 NIST CSF 2.0 outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D276 · M10W4D3 · Specialization literature and documentation reading

Phase: Academic month · **Optional date:** 2027-07-29 · **Domain resources:** B03 / B17

1. Day / sequence	Global Day 276 of 365; calendar-free M10W4D3; optional date 2027-07-29.
2. Exact topic	Specialization literature and documentation reading
3. Exact subtopics	Primary docs, standards, papers, advisories, known bypasses and implementation details
4. Learning objective	Learn to read primary sources instead of only tutorials
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ documentation and B14 Elastic detection rules repository for primary detection content. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Read two primary technical sources, write an annotated bibliography and extract three design decisions for your capstone. **2.** Standard (~2 h): Read two primary technical sources, write an annotated bibliography and extract three design decisions for your capstone. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Read two primary technical sources, write an annotated bibliography and extract three design decisions for your capstone. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	annotated-bibliography.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which source is authoritative and which is only commentary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Notes distinguish primary evidence from opinion and cite exact sections Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ documentation and B14 Elastic detection rules repository for primary detection content"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D277 · M10W4D4 · Independent problem-solving baseline with no walkthrough

Phase: Academic month · **Optional date:** 2027-07-30 · **Domain resources:** B04 / B02

1. Day / sequence	Global Day 277 of 365; calendar-free M10W4D4; optional date 2027-07-30.
2. Exact topic	Independent problem-solving baseline with no walkthrough
3. Exact subtopics	Unfamiliar rule/schema/configuration bug, hypothesis testing, documentation search and debugging
4. Learning objective	Measure ability to work without step-by-step guidance
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and CL09 Kubernetes security documentation, chosen according to specialization. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve one unfamiliar technical task in 90 minutes using only official docs and your error log; record every failed hypothesis. **2.** Standard (~2 h): Solve one unfamiliar technical task in 90 minutes using only official docs and your error log; record every failed hypothesis. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve one unfamiliar technical task in 90 minutes using only official docs and your error log; record every failed hypothesis. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	independent-baseline.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which documentation section resolved the ambiguity? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Task is completed or failure is analyzed with a concrete next experiment Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and CL09 Kubernetes security documentation, chosen according to specialization"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D278 · M10W4D5 · Month 10 practical examination and gate

Phase: Academic month · **Optional date:** 2027-07-31 · **Domain resources:** B04 / C20

1. Day / sequence	Global Day 278 of 365; calendar-free M10W4D5; optional date 2027-07-31.
2. Exact topic	Month 10 practical examination and gate
3. Exact subtopics	Specialization decision, detection engineering, cloud security, independent research, documentation and portfolio quality
4. Learning objective	Demonstrate intermediate specialization competence and decision maturity
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and CL03 Well-Architected Security Pillar as primary references. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical in the chosen primary: solve an unfamiliar technical problem, write documentation and defend the specialization decision. **2.** Standard (~2 h): Complete a 150-minute practical in the chosen primary: solve an unfamiliar technical problem, write documentation and defend the specialization decision. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical in the chosen primary: solve an unfamiliar technical problem, write documentation and defend the specialization decision. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would change your decision in the next six months? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and independent problem-solving at least 11/15 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and CL03 Well-Architected Security Pillar as primary references"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D279 · M10W4D6 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-08-01 · **Domain resources:** B03 / C17

1. Day / sequence	Global Day 279 of 365; calendar-free M10W4D6; optional date 2027-08-01.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Decision bias, telemetry gaps, query/detection quality, cloud IAM, Kubernetes controls and reading notes
4. Learning objective	Close Month 10 gaps and finalize the Month 11 capstone plan
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B14 detection rule docs; CL09 Kubernetes security docs for secondary gaps. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, update capstone spec, add ten flashcards and define a weekly milestone schedule for Month 11. **2.** Standard (~2 h): Rework failed assessment items, update capstone spec, add ten flashcards and define a weekly milestone schedule for Month 11. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, update capstone spec, add ten flashcards and define a weekly milestone schedule for Month 11. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-review.md and capstone-milestones.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which prerequisite is still too weak for the capstone and how will you fix it first? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and capstone plan has clear weekly deliverables Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B14 detection rule docs; CL09 Kubernetes security docs for secondary gaps"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

D280 · M10W4D7 · Spaced revision, recovery and Month 11 preview

Phase: Academic month · **Optional date:** 2027-08-02 · **Domain resources:** A08 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 280 of 365; calendar-free M10W4D7; optional date 2027-08-02.
2. Exact topic	Spaced revision, recovery and Month 11 preview
3. Exact subtopics	Detection lifecycle, cloud controls, research reading and decision rationale; active recall and rest
4. Learning objective	Preserve specialization gains before a month of independent engineering
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering selected chapter for long-term perspective, read only one section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, clean repositories, create the Month 11 project board, and rest or take a light walkthrough of the capstone architecture — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, clean repositories, create the Month 11 project board, and rest or take a light walkthrough of the capstone architecture — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, clean repositories, create the Month 11 project board, and rest or take a light walkthrough of the capstone architecture — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M10-consolidation.md and project-board.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What is the single riskiest assumption in your capstone plan? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Board is ready, snapshots are clean, and the risky assumption has a validation experiment Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering selected chapter for long-term perspective, read only one section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 — Capstone engineering, independent investigation and portfolio

Days D281-D308. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Capstone kickoff and problem definition D2: Capstone architecture and data-flow design D3: Threat model and abuse-case tests D4: Development environment and reproducibility D5: Build sprint 1: core application and data model D6: Build sprint 1: telemetry and evidence generation D7: Weekly capstone review 1 and design defence
Week 2	D1: Build sprint 2: automation and detection rules D2: Build sprint 2: containerization and CI/CD D3: Build sprint 2: cloud/IaC security alternative D4: Build sprint 2: failure injection and resilience D5: Build sprint 2: independent investigation exercise D6: Build sprint 2: remediation and regression testing D7: Weekly capstone review 2 and score
Week 3	D1: Independent technical challenge 1: unfamiliar log format D2: Independent technical challenge 2: unknown detection gap D3: Independent technical challenge 4: ethical decision scenario D4: Capstone security review and red-team self-test D5: Capstone remediation sprint D6: Weekly independent work review D7: Capstone technical report: architecture and methods
Week 4	D1: Capstone technical report: results and evaluation D2: Capstone report: discussion, ethics and future work D3: Capstone demo and oral defence script D4: Final portfolio integration and README polish D5: Month 11 practical examination and mastery gate D6: Monthly review and gap repairs D7: Spaced revision, recovery and Month 12 preview

> **Gate:** score the Month 11 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 11 · Week 1 · daily cards

D281 · M11W1D1 · Capstone kickoff and problem definition

Phase: Academic month · **Optional date:** 2027-08-03 · **Domain resources:** C17 / F09

1. Day / sequence	Global Day 281 of 365; calendar-free M11W1D1; optional date 2027-08-03.
2. Exact topic	Capstone kickoff and problem definition
3. Exact subtopics	Stakeholder, assets, problem, scope, constraints, success metrics, non-goals and security boundaries
4. Learning objective	Convert the Month 10 specification into a buildable engineering plan
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling community page and C06 NIST CSF 2.0 outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a project charter, milestone board, risk register and evidence plan; define what the capstone will not do. **2.** Standard (~2 h): Create a project charter, milestone board, risk register and evidence plan; define what the capstone will not do. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a project charter, milestone board, risk register and evidence plan; define what the capstone will not do. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/charter.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which non-goal prevents scope creep without making the project trivial? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Charter has measurable outcomes and explicit ethical boundaries Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling community page and C06 NIST CSF 2.0 outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D282 · M11W1D2 · Capstone architecture and data-flow design

Phase: Academic month · **Optional date:** 2027-08-04 · **Domain resources:** C15 / CL03

1. Day / sequence	Global Day 282 of 365; calendar-free M11W1D2; optional date 2027-08-04.
2. Exact topic	Capstone architecture and data-flow design
3. Exact subtopics	Components, trust boundaries, identities, data stores, APIs, logs, CI/CD, containers and recovery
4. Learning objective	Produce a security architecture that can be implemented on student hardware
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 selected architecture and logging requirements. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draw a three-tier architecture with data flows, trust boundaries, failure domains and a resource budget under 8 GB RAM where possible. **2.** Standard (~2 h): Draw a three-tier architecture with data flows, trust boundaries, failure domains and a resource budget under 8 GB RAM where possible. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draw a three-tier architecture with data flows, trust boundaries, failure domains and a resource budget under 8 GB RAM where possible. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/architecture.svg; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which component has the highest blast radius if compromised? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Architecture is feasible, labelled and has a specific isolation plan Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 selected architecture and logging requirements"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D283 · M11W1D3 · Threat model and abuse-case tests

Phase: Academic month · **Optional date:** 2027-08-05 · **Domain resources:** C17 / C26

1. Day / sequence	Global Day 283 of 365; calendar-free M11W1D3; optional date 2027-08-05.
2. Exact topic	Threat model and abuse-case tests
3. Exact subtopics	STRIDE, attack trees, misuse cases, telemetry mapping, risk ranking and mitigations
4. Learning objective	Derive testable security requirements from threats rather than a generic checklist
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	OWASP Threat Modeling and C18 MITRE ATT&CK technique pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a threat model with at least twelve threats; map each to a mitigation, test and detection data source. **2.** Standard (~2 h): Build a threat model with at least twelve threats; map each to a mitigation, test and detection data source. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a threat model with at least twelve threats; map each to a mitigation, test and detection data source. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/threat-model.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which threat has no current telemetry and how will you add it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every high/critical threat has a mitigation and validation test Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Threat Modeling and C18 MITRE ATT&CK technique pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D284 · M11W1D4 · Development environment and reproducibility

Phase: Academic month · **Optional date:** 2027-08-06 · **Domain resources:** F09 / CL14

1. Day / sequence	Global Day 284 of 365; calendar-free M11W1D4; optional date 2027-08-06.
2. Exact topic	Development environment and reproducibility
3. Exact subtopics	Repository structure, containers, dependencies, virtual environments, configuration, secrets placeholders, Makefile/scripts and clean setup
4. Learning objective	Make the capstone reproducible on a clean machine with no cloud spending
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 1-3 and CL06 Docker Get Started documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a repository skeleton, setup script, .gitignore, sample configuration and a clean-install test from scratch. **2.** Standard (~2 h): Create a repository skeleton, setup script, .gitignore, sample configuration and a clean-install test from scratch. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a repository skeleton, setup script, .gitignore, sample configuration and a clean-install test from scratch. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15 repository skeleton; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which setup step still depends on undocumented manual action? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	A second user or a fresh VM can reproduce the environment from README Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 1-3 and CL06 Docker Get Started documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D285 · M11W1D5 · Build sprint 1: core application and data model

Phase: Academic month · **Optional date:** 2027-08-07 · **Domain resources:** W17 / P08

1. Day / sequence	Global Day 285 of 365; calendar-free M11W1D5; optional date 2027-08-07.
2. Exact topic	Build sprint 1: core application and data model
3. Exact subtopics	Minimal API/service, authentication, authorization, object ownership, validation, database schema and errors
4. Learning objective	Implement the smallest trustworthy core rather than an oversized feature set
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series for Authentication, Session Management and Input Validation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement core endpoints, server-side authorization, parameterized queries, password hashing/KDF and structured error handling. **2.** Standard (~2 h): Implement core endpoints, server-side authorization, parameterized queries, password hashing/KDF and structured error handling. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement core endpoints, server-side authorization, parameterized queries, password hashing/KDF and structured error handling. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/src/ and migration files; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which input path is not yet validated and how could that fail? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Core functional tests pass and all assigned negative tests fail closed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series for Authentication, Session Management and Input Validation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D286 · M11W1D6 · Build sprint 1: telemetry and evidence generation

Phase: Academic month · **Optional date:** 2027-08-08 · **Domain resources:** B02 / W17

1. Day / sequence	Global Day 286 of 365; calendar-free M11W1D6; optional date 2027-08-08.
2. Exact topic	Build sprint 1: telemetry and evidence generation
3. Exact subtopics	Structured logs, correlation IDs, audit events, time sync, redaction, synthetic malicious/benign events and data retention
4. Learning objective	Ensure the capstone produces evidence useful for investigation
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview for schema design and B01 Sentinel data collection concepts. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add structured logging for authentication, authorization failures, privileged actions and errors; generate a labelled synthetic event dataset. **2.** Standard (~2 h): Add structured logging for authentication, authorization failures, privileged actions and errors; generate a labelled synthetic event dataset. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add structured logging for authentication, authorization failures, privileged actions and errors; generate a labelled synthetic event dataset. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/data/ and logging schema; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which privacy-sensitive field must never appear in logs? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Logs are structured, redacted and include reproducible benign/malicious examples Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview for schema design and B01 Sentinel data collection concepts"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D287 · M11W1D7 · Weekly capstone review 1 and design defence

Phase: Academic month · **Optional date:** 2027-08-09 · **Domain resources:** C15 / F09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 287 of 365; calendar-free M11W1D7; optional date 2027-08-09.
2. Exact topic	Weekly capstone review 1 and design defence
3. Exact subtopics	Charter, architecture, threat model, reproducibility, core implementation and telemetry; peer-style review
4. Learning objective	Catch design errors before deeper implementation
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 and C17 threat model review. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Conduct a 45-minute self-design review, create ten issues, fix the highest three, and record a short architecture walkthrough — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Conduct a 45-minute self-design review, create ten issues, fix the highest three, and record a short architecture walkthrough — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Conduct a 45-minute self-design review, create ten issues, fix the highest three, and record a short architecture walkthrough — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11W1-review.md and issue list; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which design change improved security without adding complexity? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All high-priority issues have owners and tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 and C17 threat model review"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 · Week 2 · daily cards

D288 · M11W2D1 · Build sprint 2: automation and detection rules

Phase: Academic month · **Optional date:** 2027-08-10 · **Domain resources:** B04 / B02

1. Day / sequence	Global Day 288 of 365; calendar-free M11W2D1; optional date 2027-08-10.
2. Exact topic	Build sprint 2: automation and detection rules
3. Exact subtopics	Log normalization, Sigma/KQL-style rules, true/false-positive tests, alert metadata and measurement
4. Learning objective	Integrate the primary specialization capability into the capstone
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started guide and B14 Elastic detection rules documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement three detections with labelled test data, CI validation and a short design note for each. **2.** Standard (~2 h): Implement three detections with labelled test data, CI validation and a short design note for each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement three detections with labelled test data, CI validation and a short design note for each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/detections/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which detection has the highest false-positive risk and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rules validate, tests pass and all false-positive cases are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started guide and B14 Elastic detection rules documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D289 · M11W2D2 · Build sprint 2: containerization and CI/CD

Phase: Academic month · **Optional date:** 2027-08-11 · **Domain resources:** CL14 / W13

1. Day / sequence	Global Day 289 of 365; calendar-free M11W2D2; optional date 2027-08-11.
2. Exact topic	Build sprint 2: containerization and CI/CD
3. Exact subtopics	Dockerfile, non-root user, image scanning, GitHub Actions, least permissions, pinned actions, tests and SBOM
4. Learning objective	Build a reproducible, secure local delivery pipeline
5. Primary lecture / reading + direct URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	GitHub Actions security hardening and CL12 Trivy documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Containerize the app, harden the Dockerfile, scan the image, add CI tests and generate an SBOM-like artifact inventory. **2.** Standard (~2 h): Containerize the app, harden the Dockerfile, scan the image, add CI tests and generate an SBOM-like artifact inventory. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Containerize the app, harden the Dockerfile, scan the image, add CI tests and generate an SBOM-like artifact inventory. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/Dockerfile and workflow evidence; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which CI secret could leak through a fork pull request? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Pipeline runs locally or in CI and no secret appears in logs Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub Actions security hardening and CL12 Trivy documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup GitHub code security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D290 · M11W2D3 · Build sprint 2: cloud/IaC security alternative

Phase: Academic month · **Optional date:** 2027-08-12 · **Domain resources:** CL15 / CL09

1. Day / sequence	Global Day 290 of 365; calendar-free M11W2D3; optional date 2027-08-12.
2. Exact topic	Build sprint 2: cloud/IaC security alternative
3. Exact subtopics	Synthetic Terraform/Kubernetes manifests, IAM/network policies, Checkov checks and no live deployment
4. Learning objective	Integrate infrastructure security without creating cloud costs
5. Primary lecture / reading + direct URL	Terraform documentation `https://developer.hashicorp.com/terraform/docs` **Type:** Official documentation · **Access:** Free/open-source; HCP Terraform has paid tiers
6. Exact section / lesson locator	Terraform documentation and CL13 Checkov documentation; CL03 Security Pillar for controls. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Add a local IaC representation of the capstone, scan it, fix five findings and document the shared-responsibility boundary. **2.** Standard (~2 h): Add a local IaC representation of the capstone, scan it, fix five findings and document the shared-responsibility boundary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Add a local IaC representation of the capstone, scan it, fix five findings and document the shared-responsibility boundary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/iac/ and scan report; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control would still be the customer's responsibility in a managed service? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No billable resource is deployed and scan findings are triaged Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Terraform documentation and CL13 Checkov documentation; CL03 Security Pillar for controls"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Checkov documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D291 · M11W2D4 · Build sprint 2: failure injection and resilience

Phase: Academic month · **Optional date:** 2027-08-13 · **Domain resources:** C27 / P05

1. Day / sequence	Global Day 291 of 365; calendar-free M11W2D4; optional date 2027-08-13.
2. Exact topic	Build sprint 2: failure injection and resilience
3. Exact subtopics	Service failure, malformed log, database lock, missing permission, expired credential, network partition and restore
4. Learning objective	Prove the capstone fails safely and can be recovered
5. Primary lecture / reading + direct URL	NIST SP 800-34 Revision 1 `https://csrc.nist.gov/pubs/sp/800/34/r1/final` **Type:** Official continuity guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-34r1 backup, recovery and testing sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create six failure-injection tests, measure detection and recovery, fix two failures and document residual risk. **2.** Standard (~2 h): Create six failure-injection tests, measure detection and recovery, fix two failures and document residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create six failure-injection tests, measure detection and recovery, fix two failures and document residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/tests/failure-injection.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which failure was silent and how did you add an observable signal? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests are reproducible and at least two fixes are verified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-34r1 backup, recovery and testing sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D292 · M11W2D5 · Build sprint 2: independent investigation exercise

Phase: Academic month · **Optional date:** 2027-08-14 · **Domain resources:** B15 / B02

1. Day / sequence	Global Day 292 of 365; calendar-free M11W2D5; optional date 2027-08-14.
2. Exact topic	Build sprint 2: independent investigation exercise
3. Exact subtopics	Unknown alert, incomplete logs, competing hypotheses, pivoting, confidence and evidence gaps
4. Learning objective	Investigate your own system without using a known answer
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide, hunt/investigation sections; C07 NIST SP 800-61r3. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Ask another person or use a script to inject one unknown benign/malicious event; investigate from logs only and write a case file. **2.** Standard (~2 h): Ask another person or use a script to inject one unknown benign/malicious event; investigate from logs only and write a case file. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Ask another person or use a script to inject one unknown benign/malicious event; investigate from logs only and write a case file. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/investigation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which hypothesis was rejected by a specific log field? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Case file includes evidence, timeline, uncertainty and a conclusion Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide, hunt/investigation sections; C07 NIST SP 800-61r3"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D293 · M11W2D6 · Build sprint 2: remediation and regression testing

Phase: Academic month · **Optional date:** 2027-08-15 · **Domain resources:** W17 / B03

1. Day / sequence	Global Day 293 of 365; calendar-free M11W2D6; optional date 2027-08-15.
2. Exact topic	Build sprint 2: remediation and regression testing
3. Exact subtopics	Root-cause fixes, rule tuning, new tests, rollback, functional regression and documentation
4. Learning objective	Close the investigation loop with engineering changes rather than a report alone
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series relevant to the finding and P05 pytest docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Implement fixes for the highest finding, add regression and detection tests, rerun full suite and record before/after metrics. **2.** Standard (~2 h): Implement fixes for the highest finding, add regression and detection tests, rerun full suite and record before/after metrics. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Implement fixes for the highest finding, add regression and detection tests, rerun full suite and record before/after metrics. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/remediation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which test proves the fix did not only block one payload? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests pass and the report distinguishes root cause from symptom Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series relevant to the finding and P05 pytest docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D294 · M11W2D7 · Weekly capstone review 2 and score

Phase: Academic month · **Optional date:** 2027-08-16 · **Domain resources:** B03 / CL14 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 294 of 365; calendar-free M11W2D7; optional date 2027-08-16.
2. Exact topic	Weekly capstone review 2 and score
3. Exact subtopics	Automation, detection, containers, CI/CD, IaC, resilience, investigation and remediation; active recall
4. Learning objective	Evaluate progress against the capstone rubric and adjust Month 11 Week 3 scope
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B14 detection rule documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub Actions security hardening `https://docs.github.com/en/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the current capstone with the 100-point rubric, identify three largest gaps and replan the remaining work with explicit cut lines — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the current capstone with the 100-point rubric, identify three largest gaps and replan the remaining work with explicit cut lines — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Score the current capstone with the 100-point rubric, identify three largest gaps and replan the remaining work with explicit cut lines — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11W2-review.md and updated board; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which feature should be cut to protect quality? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scope is reduced where necessary and evidence gaps have owners Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B14 detection rule documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 · Week 3 · daily cards

D295 · M11W3D1 · Independent technical challenge 1: unfamiliar log format

Phase: Academic month · **Optional date:** 2027-08-17 · **Domain resources:** F08 / P03

1. Day / sequence	Global Day 295 of 365; calendar-free M11W3D1; optional date 2027-08-17.
2. Exact topic	Independent technical challenge 1: unfamiliar log format
3. Exact subtopics	Schema discovery, parsing, normalization, edge cases and validation without a walkthrough
4. Learning objective	Demonstrate independent problem-solving on an unknown but safe dataset
5. Primary lecture / reading + direct URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Python Tutorial files/exceptions sections and P04 RegexOne review lessons. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Automate the Boring Stuff with Python `https://automatetheboringstuff.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Receive or generate an unseen synthetic log format; write a parser and tests in 90 minutes using only docs and your notes. **2.** Standard (~2 h): Receive or generate an unseen synthetic log format; write a parser and tests in 90 minutes using only docs and your notes. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Receive or generate an unseen synthetic log format; write a parser and tests in 90 minutes using only docs and your notes. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	challenge1/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which assumption about time or encoding was wrong and how did you find it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Parser works on all test cases without copied code Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Python Tutorial files/exceptions sections and P04 RegexOne review lessons"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Learn Python - Full Course for Beginners (freeCodeCamp); write a five-line correction; retest next morning with a new 10-minute problem.

D296 · M11W3D2 · Independent technical challenge 2: unknown detection gap

Phase: Academic month · **Optional date:** 2027-08-18 · **Domain resources:** C18 / B04

1. Day / sequence	Global Day 296 of 365; calendar-free M11W3D2; optional date 2027-08-18.
2. Exact topic	Independent technical challenge 2: unknown detection gap
3. Exact subtopics	Behaviour hypothesis, telemetry requirements, rule, test data, false positives and measurement
4. Learning objective	Design a detection for a behaviour not covered by your existing rules
5. Primary lecture / reading + direct URL	MITRE ATT&CK `https://attack.mitre.org/` **Type:** Knowledge base · **Access:** Free
6. Exact section / lesson locator	MITRE ATT&CK technique page chosen from your threat model. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Pick one uncovered technique, implement a detection, create ten labelled events and measure results; explain one bypass. **2.** Standard (~2 h): Pick one uncovered technique, implement a detection, create ten labelled events and measure results; explain one bypass. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Pick one uncovered technique, implement a detection, create ten labelled events and measure results; explain one bypass. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	challenge2/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which missing field would make the detection impossible? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule is original, tested and has explicit limitations Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "MITRE ATT&CK technique page chosen from your threat model"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE D3FEND; write a five-line correction; retest next morning with a new 10-minute problem.

D297 · M11W3D3 · Independent technical challenge 4: ethical decision scenario

Phase: Academic month · **Optional date:** 2027-08-19 · **Domain resources:** E01 / C24

1. Day / sequence	Global Day 297 of 365; calendar-free M11W3D3; optional date 2027-08-19.
2. Exact topic	Independent technical challenge 4: ethical decision scenario
3. Exact subtopics	Scope ambiguity, privacy, legal constraints, business impact, safety and communication
4. Learning objective	Make and defend a decision when technical ability conflicts with authorization
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115 planning and ethical considerations; E05 HackerOne Disclosure Guidelines. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze five grey-area scenarios, choose stop/continue/escalate for each and write the exact message to the stakeholder. **2.** Standard (~2 h): Analyze five grey-area scenarios, choose stop/continue/escalate for each and write the exact message to the stakeholder. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze five grey-area scenarios, choose stop/continue/escalate for each and write the exact message to the stakeholder. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	ethics-scenarios.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which action is technically easy but professionally unacceptable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All decisions prioritize authorization, privacy and availability Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115 planning and ethical considerations; E05 HackerOne Disclosure Guidelines"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D298 · M11W3D4 · Capstone security review and red-team self-test

Phase: Academic month · **Optional date:** 2027-08-20 · **Domain resources:** W06 / C17

1. Day / sequence	Global Day 298 of 365; calendar-free M11W3D4; optional date 2027-08-20.
2. Exact topic	Capstone security review and red-team self-test
3. Exact subtopics	Authorized local assessment, abuse cases, finding validation, evidence, impact and false positives
4. Learning objective	Test your own capstone as an ethical assessor
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP Web Security Testing Guide and E01 NIST SP 800-115 validation sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a scoped assessment against the local capstone, record five findings or rejected hypotheses, and preserve evidence. **2.** Standard (~2 h): Run a scoped assessment against the local capstone, record five findings or rejected hypotheses, and preserve evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a scoped assessment against the local capstone, record five findings or rejected hypotheses, and preserve evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/assessment/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding surprised you and what design assumption caused it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All tests remain local/authorized and findings are reproducible Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Web Security Testing Guide and E01 NIST SP 800-115 validation sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D299 · M11W3D5 · Capstone remediation sprint

Phase: Academic month · **Optional date:** 2027-08-21 · **Domain resources:** W17 / P05

1. Day / sequence	Global Day 299 of 365; calendar-free M11W3D5; optional date 2027-08-21.
2. Exact topic	Capstone remediation sprint
3. Exact subtopics	Root-cause fixes, compensating controls, tests, rollback, residual risk and retest
4. Learning objective	Close high-priority findings before final reporting
5. Primary lecture / reading + direct URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/` **Type:** Official checklist library · **Access:** Free
6. Exact section / lesson locator	OWASP Cheat Sheet Series relevant to findings. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Fix the three highest findings, write regression tests, rerun the assessment and document any accepted residual risk. **2.** Standard (~2 h): Fix the three highest findings, write regression tests, rerun the assessment and document any accepted residual risk. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Fix the three highest findings, write regression tests, rerun the assessment and document any accepted residual risk. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/remediation-report.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What evidence proves the root cause was fixed? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All high-priority findings are retested and accepted risks have owner and review date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Cheat Sheet Series relevant to findings"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Application Security Verification Standard; write a five-line correction; retest next morning with a new 10-minute problem.

D300 · M11W3D6 · Weekly independent work review

Phase: Academic month · **Optional date:** 2027-08-22 · **Domain resources:** C20 / B03

1. Day / sequence	Global Day 300 of 365; calendar-free M11W3D6; optional date 2027-08-22.
2. Exact topic	Weekly independent work review
3. Exact subtopics	Four challenges, security review and remediation; peer-style scoring
4. Learning objective	Demonstrate Month 11 Week 3 competence and identify remaining report gaps
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks for the chosen specialization. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Score all four challenges with the 100-point rubric, choose the strongest as portfolio evidence and list three final improvements. **2.** Standard (~2 h): Score all four challenges with the 100-point rubric, choose the strongest as portfolio evidence and list three final improvements. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Score all four challenges with the 100-point rubric, choose the strongest as portfolio evidence and list three final improvements. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11W3-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which challenge best demonstrates independent capability? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scores are evidence-based and the chosen artifact is linked in the portfolio plan Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks for the chosen specialization"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D301 · M11W3D7 · Capstone technical report: architecture and methods

Phase: Academic month · **Optional date:** 2027-08-23 · **Domain resources:** A08 / F09 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 301 of 365; calendar-free M11W3D7; optional date 2027-08-23.
2. Exact topic	Capstone technical report: architecture and methods
3. Exact subtopics	Problem, related work/primary sources, architecture, data, methods, assumptions, threat model and reproducibility
4. Learning objective	Write a technical report that another student could replicate
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering selected chapter on technical communication and A12 USENIX Security paper structure examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Draft the first six sections of the final capstone report with diagrams, exact setup instructions and citations to primary docs — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Draft the first six sections of the final capstone report with diagrams, exact setup instructions and citations to primary docs — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Draft the first six sections of the final capstone report with diagrams, exact setup instructions and citations to primary docs — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/report-draft.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which implementation decision needs a clear rationale? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report is reproducible and distinguishes your work from cited work Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering selected chapter on technical communication and A12 USENIX Security paper structure examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Month 11 · Week 4 · daily cards

D302 · M11W4D1 · Capstone technical report: results and evaluation

Phase: Academic month · **Optional date:** 2027-08-24 · **Domain resources:** B14 / C26

1. Day / sequence	Global Day 302 of 365; calendar-free M11W4D1; optional date 2027-08-24.
2. Exact topic	Capstone technical report: results and evaluation
3. Exact subtopics	Functional results, security tests, detection metrics, failure injection, performance, limitations and negative results
4. Learning objective	Present evidence honestly, including failures
5. Primary lecture / reading + direct URL	Elastic detection rules documentation `https://www.elastic.co/guide/en/security/current/detection-engine-overview.html` **Type:** Official docs / open-source rules · **Access:** Free docs/open-source rules; Elastic Stack may have paid features
6. Exact section / lesson locator	Elastic detection rules docs for measurement concepts and P05 pytest reporting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-30 Revision 1 `https://csrc.nist.gov/pubs/sp/800/30/r1/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build tables/charts from real capstone test outputs; report false positives, missed cases, resource use and limitations honestly. **2.** Standard (~2 h): Build tables/charts from real capstone test outputs; report false positives, missed cases, resource use and limitations honestly. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build tables/charts from real capstone test outputs; report false positives, missed cases, resource use and limitations honestly. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/results.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which result did not match your hypothesis and what did you learn? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All numbers come from reproducible tests and no success is exaggerated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Elastic detection rules docs for measurement concepts and P05 pytest reporting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D303 · M11W4D2 · Capstone report: discussion, ethics and future work

Phase: Academic month · **Optional date:** 2027-08-25 · **Domain resources:** C24 / C07

1. Day / sequence	Global Day 303 of 365; calendar-free M11W4D2; optional date 2027-08-25.
2. Exact topic	Capstone report: discussion, ethics and future work
3. Exact subtopics	Impact, generalizability, ethical boundaries, privacy, operational constraints, cost and future experiments
4. Learning objective	Defend the capstone's value without claiming more than it proves
5. Primary lecture / reading + direct URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj` **Type:** Regulation/legal text · **Access:** Free
6. Exact section / lesson locator	GDPR privacy-by-design principles and E05 disclosure guidance. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write discussion and future-work sections, explicitly state authorization, synthetic data use and limitations of a student lab. **2.** Standard (~2 h): Write discussion and future-work sections, explicitly state authorization, synthetic data use and limitations of a student lab. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write discussion and future-work sections, explicitly state authorization, synthetic data use and limitations of a student lab. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/discussion.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would a production environment reveal that your lab cannot? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Section is candid and avoids unrealistic claims Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GDPR privacy-by-design principles and E05 disclosure guidance"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Privacy Framework; write a five-line correction; retest next morning with a new 10-minute problem.

D304 · M11W4D3 · Capstone demo and oral defence script

Phase: Academic month · **Optional date:** 2027-08-26 · **Domain resources:** C17 / B03

1. Day / sequence	Global Day 304 of 365; calendar-free M11W4D3; optional date 2027-08-26.
2. Exact topic	Capstone demo and oral defence script
3. Exact subtopics	Ten-minute demo, architecture walkthrough, threat model, live test, failure, detection, remediation and Q&A
4. Learning objective	Prepare to explain and defend technical choices under questioning
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	Threat model and C15 ASVS requirements as review checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Record or rehearse a ten-minute demo; prepare answers to ten probable reviewer questions with evidence links. **2.** Standard (~2 h): Record or rehearse a ten-minute demo; prepare answers to ten probable reviewer questions with evidence links. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Record or rehearse a ten-minute demo; prepare answers to ten probable reviewer questions with evidence links. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	PRJ-15/demo-script.md and recording notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which question exposes the weakest evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Demo works from a clean setup and all answers cite specific files/tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Threat model and C15 ASVS requirements as review checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D305 · M11W4D4 · Final portfolio integration and README polish

Phase: Academic month · **Optional date:** 2027-08-27 · **Domain resources:** F09 / C20

1. Day / sequence	Global Day 305 of 365; calendar-free M11W4D4; optional date 2027-08-27.
2. Exact topic	Final portfolio integration and README polish
3. Exact subtopics	Repository structure, setup, usage, examples, screenshots, reports, limitations, licence, changelog and release tag
4. Learning objective	Make every Month 11 artifact discoverable and reproducible
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and W13 GitHub security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Update the portfolio index, link PRJ-01 through PRJ-15, sanitize all evidence and tag a release candidate. **2.** Standard (~2 h): Update the portfolio index, link PRJ-01 through PRJ-15, sanitize all evidence and tag a release candidate. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Update the portfolio index, link PRJ-01 through PRJ-15, sanitize all evidence and tag a release candidate. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	portfolio index and release tag; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which public file still contains personal data or a secret-like string? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Automated/manual secret scan passes and all links work Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and W13 GitHub security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D306 · M11W4D5 · Month 11 practical examination and mastery gate

Phase: Academic month · **Optional date:** 2027-08-28 · **Domain resources:** C17 / B03

1. Day / sequence	Global Day 306 of 365; calendar-free M11W4D5; optional date 2027-08-28.
2. Exact topic	Month 11 practical examination and mastery gate
3. Exact subtopics	Capstone design, implementation, security, detection, incident response, independent challenges and technical writing
4. Learning objective	Demonstrate advanced foundational competence and independent engineering
5. Primary lecture / reading + direct URL	OWASP Threat Modeling community page `https://owasp.org/www-community/Threat_Modeling` **Type:** Reference/process · **Access:** Free
6. Exact section / lesson locator	Threat model and C15 ASVS Level 1 review checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 180-minute capstone defence: implement or fix one unseen requirement, investigate a new event, update the report and answer oral questions. **2.** Standard (~2 h): Complete a 180-minute capstone defence: implement or fix one unseen requirement, investigate a new event, update the report and answer oral questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 180-minute capstone defence: implement or fix one unseen requirement, investigate a new event, update the report and answer oral questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11-assessment.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which part of your capstone is original rather than assembled from tutorials? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 75/100 and independent problem-solving at least 12/15 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Threat model and C15 ASVS Level 1 review checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup MITRE ATT&CK; write a five-line correction; retest next morning with a new 10-minute problem.

D307 · M11W4D6 · Monthly review and gap repairs

Phase: Academic month · **Optional date:** 2027-08-29 · **Domain resources:** P05 / F09

1. Day / sequence	Global Day 307 of 365; calendar-free M11W4D6; optional date 2027-08-29.
2. Exact topic	Monthly review and gap repairs
3. Exact subtopics	Capstone architecture, evidence, tests, reports, demo and portfolio; active recall and issue triage
4. Learning objective	Close Month 11 gaps before final Month 12 examinations
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest docs and B04 Sigma guide for weak tests/rules. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rework failed assessment items, fix top five portfolio issues, update the capstone report and create a final revision checklist. **2.** Standard (~2 h): Rework failed assessment items, fix top five portfolio issues, update the capstone report and create a final revision checklist. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rework failed assessment items, fix top five portfolio issues, update the capstone report and create a final revision checklist. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11-review.md and final-issues.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which gap could cause failure in the Month 12 practical? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and all critical issues are closed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest docs and B04 Sigma guide for weak tests/rules"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D308 · M11W4D7 · Spaced revision, recovery and Month 12 preview

Phase: Academic month · **Optional date:** 2027-08-30 · **Domain resources:** B03 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 308 of 365; calendar-free M11W4D7; optional date 2027-08-30.
2. Exact topic	Spaced revision, recovery and Month 12 preview
3. Exact subtopics	Capstone decisions, detection metrics, cloud controls, secure coding and report claims; light retrieval
4. Learning objective	Preserve Month 11 gains and prepare final diagnostic exams
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and CL03 Security Pillar review pages. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, take a full clean snapshot of the capstone, archive evidence, review Month 12 exam rules and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, take a full clean snapshot of the capstone, archive evidence, review Month 12 exam rules and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do a 45-minute retrieval session, take a full clean snapshot of the capstone, archive evidence, review Month 12 exam rules and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M11-consolidation.md and clean snapshot log; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which topic from Months 1-9 has decayed most and needs first revision? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Capstone is backed up, no active lab is exposed, and revision order is set Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and CL03 Security Pillar review pages"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 — Practical examinations, revision, portfolio and long-term planning

Days D309-D336. Each day has the same 13-field structure. Use the Core, Standard or Intensive track; the learning objective does not change. Day 7 of each week is lighter by design.

Week	Daily topics
Week 1	D1: Final diagnostic: foundations and networking D2: Gap repair 1: computer systems and OS D3: Gap repair 2: networking and packet analysis D4: Gap repair 3: Linux, Windows and scripting D5: Weekly diagnostic review and revision plan D6: Gap repair 4: core security and GRC D7: Gap repair 5: web and API security
Week 2	D1: Gap repair 6: ethical assessment and blue team D2: Gap repair 7: cloud and infrastructure security D3: Mock practical examination 1: guided recall D4: Mock practical examination 2: independent conditions D5: Weekly review of mocks and targeted repair D6: Final practical examination part 1: systems and network D7: Final practical examination part 2: web and ethical assessment
Week 3	D1: Final practical examination part 3: blue team and cloud D2: Final capstone defence and oral examination D3: Weekly final examination review and grade calculation D4: Final portfolio audit and sanitization D5: CV, LinkedIn and technical story preparation D6: Technical interview practice and scenario answers D7: Certification readiness and final decision
Week 4	D1: Year 2-5 long-term plan and 90-day sprint D2: Monthly review, celebration and responsible handover D3: Master resource directory update and link re-verification D4: Practice-exam strategy and question analysis D5: Internship and entry-level application strategy D6: Sustainable study system and burnout prevention D7: Spaced revision and handover to final consolidation

> **Gate:** score the Month 12 assessment using the 100-point rubric in Volume 6. If score <70 or safety <9/10, follow the six-step repair protocol before advancing.

Month 12 · Week 1 · daily cards

D309 · M12W1D1 · Final diagnostic: foundations and networking

Phase: Academic month · **Optional date:** 2027-08-31 · **Domain resources:** F01 / N16

1. Day / sequence	Global Day 309 of 365; calendar-free M12W1D1; optional date 2027-08-31.
2. Exact topic	Final diagnostic: foundations and networking
3. Exact subtopics	Computer architecture, OS, CLI, HTTP, SQL, Git, OSI/TCPIP, subnetting, TCP/UDP, DNS, TLS and packet analysis
4. Learning objective	Measure current competence and identify the highest-risk knowledge gaps
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x 2026 Weeks 0-9 review pages; N01 Jeremy's IT Lab CCNA Course Index Days 1-38. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Subnetting Practice `https://subnettingpractice.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 90-minute closed-book diagnostic with 60 questions, 20 subnetting problems and one short packet analysis; classify errors by prerequisite. **2.** Standard (~2 h): Complete a 90-minute closed-book diagnostic with 60 questions, 20 subnetting problems and one short packet analysis; classify errors by prerequisite. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 90-minute closed-book diagnostic with 60 questions, 20 subnetting problems and one short packet analysis; classify errors by prerequisite. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-diagnostic-foundations.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which wrong answer reveals a conceptual error rather than a memory lapse? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All errors are classified and a revision queue is created Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x 2026 Weeks 0-9 review pages; N01 Jeremy's IT Lab CCNA Course Index Days 1-38"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D310 · M12W1D2 · Gap repair 1: computer systems and OS

Phase: Academic month · **Optional date:** 2027-09-01 · **Domain resources:** F06 / L01

1. Day / sequence	Global Day 310 of 365; calendar-free M12W1D2; optional date 2027-09-01.
2. Exact topic	Gap repair 1: computer systems and OS
3. Exact subtopics	Process/thread, memory, permissions, filesystems, services, virtualization and troubleshooting
4. Learning objective	Restore Month 1 competence to at least the passing threshold
5. Primary lecture / reading + direct URL	Operating Systems: Three Easy Pieces (OSTEP) `https://pages.cs.wisc.edu/~remzi/OSTEP/` **Type:** Free textbook · **Access:** Free online chapters; paid print optional
6. Exact section / lesson locator	OSTEP selected chapters on processes, memory and persistence. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Linux Journey `https://linuxjourney.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Revisit failed diagnostic topics, perform three practical repairs on a clean VM and retest with new questions. **2.** Standard (~2 h): Revisit failed diagnostic topics, perform three practical repairs on a clean VM and retest with new questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Revisit failed diagnostic topics, perform three practical repairs on a clean VM and retest with new questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-OS.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which OS mechanism explains two seemingly unrelated errors? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent with practical evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OSTEP selected chapters on processes, memory and persistence"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x 2026 OpenCourseWare; write a five-line correction; retest next morning with a new 10-minute problem.

D311 · M12W1D3 · Gap repair 2: networking and packet analysis

Phase: Academic month · **Optional date:** 2027-09-02 · **Domain resources:** N01 / N07

1. Day / sequence	Global Day 311 of 365; calendar-free M12W1D3; optional date 2027-09-02.
2. Exact topic	Gap repair 2: networking and packet analysis
3. Exact subtopics	Encapsulation, ARP, VLANs, IPv4/IPv6, CIDR, routing, NAT, TCP, UDP, DNS, DHCP, TLS and Wireshark
4. Learning objective	Restore Month 2 competence and daily subnetting fluency
5. Primary lecture / reading + direct URL	Jeremy's IT Lab Free CCNA 200-301 Complete Course `https://www.youtube.com/playlist?list=PLxbwE86jKRgMpuZuLBivzlM8s2Dk5lXBQ` **Type:** YouTube playlist · **Access:** Free on YouTube; optional paid course extras
6. Exact section / lesson locator	Jeremy's IT Lab CCNA Course Index selected Day titles from error log; N06 Wireshark User's Guide. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rebuild one small Packet Tracer topology, analyze one unknown sample capture, and complete 30 subnetting questions. **2.** Standard (~2 h): Rebuild one small Packet Tracer topology, analyze one unknown sample capture, and complete 30 subnetting questions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rebuild one small Packet Tracer topology, analyze one unknown sample capture, and complete 30 subnetting questions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-networking.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which packet field disproves your first hypothesis? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and capture analysis is packet-backed Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Jeremy's IT Lab CCNA Course Index selected Day titles from error log; N06 Wireshark User's Guide"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Jeremy's IT Lab CCNA Course Index and Files; write a five-line correction; retest next morning with a new 10-minute problem.

D312 · M12W1D4 · Gap repair 3: Linux, Windows and scripting

Phase: Academic month · **Optional date:** 2027-09-03 · **Domain resources:** L01 / P05

1. Day / sequence	Global Day 312 of 365; calendar-free M12W1D4; optional date 2027-09-03.
2. Exact topic	Gap repair 3: Linux, Windows and scripting
3. Exact subtopics	Linux permissions/services, Windows events/PowerShell/AD, Python, Bash, PowerShell, Git and testing
4. Learning objective	Restore Month 3-4 competence with hands-on evidence
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey review sections; L08 PowerShell documentation; F08 Python Tutorial. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	pytest documentation `https://docs.pytest.org/en/stable/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Fix one broken Python script, one Bash script and one PowerShell script; add tests and explain the root cause of each. **2.** Standard (~2 h): Fix one broken Python script, one Bash script and one PowerShell script; add tests and explain the root cause of each. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Fix one broken Python script, one Bash script and one PowerShell script; add tests and explain the root cause of each. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-scripting.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which bug was caused by platform-specific assumptions? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All three scripts pass regression tests and explanations are specific Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey review sections; L08 PowerShell documentation; F08 Python Tutorial"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D313 · M12W1D5 · Weekly diagnostic review and revision plan

Phase: Academic month · **Optional date:** 2027-09-04 · **Domain resources:** C20 / F09

1. Day / sequence	Global Day 313 of 365; calendar-free M12W1D5; optional date 2027-09-04.
2. Exact topic	Weekly diagnostic review and revision plan
3. Exact subtopics	Diagnostic results, gap repairs, error patterns, time budget and final exam strategy
4. Learning objective	Consolidate Month 12 Week 1 and schedule remaining revision
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks for competence mapping. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a heat map of all 11 modules, rank the ten weakest topics and allocate the remaining 21 days with catch-up buffers. **2.** Standard (~2 h): Create a heat map of all 11 modules, rank the ten weakest topics and allocate the remaining 21 days with catch-up buffers. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a heat map of all 11 modules, rank the ten weakest topics and allocate the remaining 21 days with catch-up buffers. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-revision-plan.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which two weak topics could cause cascading failure in the final practical? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is realistic for the chosen 1/2/3-4 hour track Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks for competence mapping"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D314 · M12W1D6 · Gap repair 4: core security and GRC

Phase: Academic month · **Optional date:** 2027-09-05 · **Domain resources:** C06 / C03

1. Day / sequence	Global Day 314 of 365; calendar-free M12W1D6; optional date 2027-09-05.
2. Exact topic	Gap repair 4: core security and GRC
3. Exact subtopics	CIA/AAA, risk, crypto, IAM, TLS, controls, IR, privacy, NIST CSF, ISO/GDPR distinctions and BCDR
4. Learning objective	Restore Month 5 competence using scenarios rather than definitions
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 https://www.nist.gov/cyberframework **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 and C26 NIST SP 800-30 risk assessment sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISC2 Certified in Cybersecurity Exam Outline https://www.isc2.org/certifications/cc/cc-certification-exam-outline
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Solve an integrated risk/IR scenario, select cryptographic primitives, classify controls and write a risk register. **2.** Standard (~2 h): Solve an integrated risk/IR scenario, select cryptographic primitives, classify controls and write a risk register. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Solve an integrated risk/IR scenario, select cryptographic primitives, classify controls and write a risk register. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-core-security.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which control category is missing from your first answer? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and all framework classifications are correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 and C26 NIST SP 800-30 risk assessment sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D315 · M12W1D7 · Gap repair 5: web and API security

Phase: Academic month · **Optional date:** 2027-09-06 · **Domain resources:** W01 / W06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 315 of 365; calendar-free M12W1D7; optional date 2027-09-06.
2. Exact topic	Gap repair 5: web and API security
3. Exact subtopics	HTTP, sessions, access control, SQLi, XSS, CSRF, SSRF, file upload, API BOLA, rate limits, secrets, SDLC and supply chain
4. Learning objective	Restore Month 6 competence with defensive fixes
5. Primary lecture / reading + direct URL	PortSwigger Web Security Academy `https://portswigger.net/web-security` **Type:** Free interactive academy · **Access:** Free account required
6. Exact section / lesson locator	PortSwigger Web Security Academy all-labs index; W17 OWASP Cheat Sheet Series. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve two unseen Apprentice labs and fix three vulnerable local code snippets with regression tests — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve two unseen Apprentice labs and fix three vulnerable local code snippets with regression tests — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Solve two unseen Apprentice labs and fix three vulnerable local code snippets with regression tests — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-web.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which fix addresses the root cause and which only blocks one payload? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and all fixes have tests Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "PortSwigger Web Security Academy all-labs index; W17 OWASP Cheat Sheet Series"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy Learning Paths; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 · Week 2 · daily cards

D316 · M12W2D1 · Gap repair 6: ethical assessment and blue team

Phase: Academic month · **Optional date:** 2027-09-07 · **Domain resources:** E01 / B03

1. Day / sequence	Global Day 316 of 365; calendar-free M12W2D1; optional date 2027-09-07.
2. Exact topic	Gap repair 6: ethical assessment and blue team
3. Exact subtopics	Authorization, scope, discovery, validation, evidence, CVSS, reporting, SOC workflow, KQL, Sigma, ATT&CK, hunting and IR
4. Learning objective	Restore Months 7-8 competence through a mixed investigation
5. Primary lecture / reading + direct URL	NIST SP 800-115 `https://csrc.nist.gov/pubs/sp/800/115/final` **Type:** Official testing guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-115; B04 Sigma getting-started; B02 KQL overview. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: In a synthetic case, write scope boundaries, validate two findings, investigate logs, write one detection rule and an incident summary. **2.** Standard (~2 h): In a synthetic case, write scope boundaries, validate two findings, investigate logs, write one detection rule and an incident summary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): In a synthetic case, write scope boundaries, validate two findings, investigate logs, write one detection rule and an incident summary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-offense-defense.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding is unsupported by evidence and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and no scope/safety error remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-115; B04 Sigma getting-started; B02 KQL overview"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Web Security Testing Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D317 · M12W2D2 · Gap repair 7: cloud and infrastructure security

Phase: Academic month · **Optional date:** 2027-09-08 · **Domain resources:** CL03 / CL09

1. Day / sequence	Global Day 317 of 365; calendar-free M12W2D2; optional date 2027-09-08.
2. Exact topic	Gap repair 7: cloud and infrastructure security
3. Exact subtopics	Shared responsibility, IAM, storage, logging, Docker, Kubernetes, CI/CD, IaC, SBOM and cloud incident response
4. Learning objective	Restore Month 9 competence with local-only evidence
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar; CL14 GitHub Actions hardening. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; fix ten prioritized issues without live deployment. **2.** Standard (~2 h): Review an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; fix ten prioritized issues without live deployment. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review an unseen cloud architecture, Dockerfile, Kubernetes manifest and workflow; fix ten prioritized issues without live deployment. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-repair-cloud.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding crosses the control-plane trust boundary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Retest score at least 80 percent and no billable resource is created Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar; CL14 GitHub Actions hardening"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D318 · M12W2D3 · Mock practical examination 1: guided recall

Phase: Academic month · **Optional date:** 2027-09-09 · **Domain resources:** C20 / B04

1. Day / sequence	Global Day 318 of 365; calendar-free M12W2D3; optional date 2027-09-09.
2. Exact topic	Mock practical examination 1: guided recall
3. Exact subtopics	All modules; time management, question interpretation, evidence and safety
4. Learning objective	Establish a baseline under exam conditions with feedback allowed afterward
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework tasks and C06 CSF 2.0 outcomes. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 120-minute mixed practical with networking, scripting, web, detection and cloud tasks; self-score with the 100-point rubric. **2.** Standard (~2 h): Complete a 120-minute mixed practical with networking, scripting, web, detection and cloud tasks; self-score with the 100-point rubric. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 120-minute mixed practical with networking, scripting, web, detection and cloud tasks; self-score with the 100-point rubric. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	mock-exam-1.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which task consumed disproportionate time and what prerequisite caused it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score is recorded with evidence and every failure has a repair action Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework tasks and C06 CSF 2.0 outcomes"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D319 · M12W2D4 · Mock practical examination 2: independent conditions

Phase: Academic month · **Optional date:** 2027-09-10 · **Domain resources:** B15 / W06

1. Day / sequence	Global Day 319 of 365; calendar-free M12W2D4; optional date 2027-09-10.
2. Exact topic	Mock practical examination 2: independent conditions
3. Exact subtopics	Unfamiliar incident, ambiguous requirements, incomplete data, independent research and technical writing
4. Learning objective	Simulate the final Month 12 practical without walkthroughs
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and C07 NIST SP 800-61r3 for reference only during open-book section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete a 150-minute practical using only primary docs; produce a report, detection, remediation and a short executive summary. **2.** Standard (~2 h): Complete a 150-minute practical using only primary docs; produce a report, detection, remediation and a short executive summary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete a 150-minute practical using only primary docs; produce a report, detection, remediation and a short executive summary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	mock-exam-2.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which assumption was unsafe to make and how did you handle it? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 70/100 and safety at least 9/10 Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and C07 NIST SP 800-61r3 for reference only during open-book section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D320 · M12W2D5 · Weekly review of mocks and targeted repair

Phase: Academic month · **Optional date:** 2027-09-11 · **Domain resources:** P05 / C20

1. Day / sequence	Global Day 320 of 365; calendar-free M12W2D5; optional date 2027-09-11.
2. Exact topic	Weekly review of mocks and targeted repair
3. Exact subtopics	Mock exam errors, time management, documentation gaps, troubleshooting and stress control
4. Learning objective	Convert mock results into a final-week plan
5. Primary lecture / reading + direct URL	pytest documentation `https://docs.pytest.org/en/stable/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	pytest documentation and B03 SigmaHQ for weak test/rule areas. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Repair the three lowest rubric categories with new exercises; rewrite one weak report and rehearse one troubleshooting scenario. **2.** Standard (~2 h): Repair the three lowest rubric categories with new exercises; rewrite one weak report and rehearse one troubleshooting scenario. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Repair the three lowest rubric categories with new exercises; rewrite one weak report and rehearse one troubleshooting scenario. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12W2-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which error is most likely to recur under pressure? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All three weak categories have new evidence and a prevention checklist Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pytest documentation and B03 SigmaHQ for weak test/rule areas"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python 3 Official Tutorial; write a five-line correction; retest next morning with a new 10-minute problem.

D321 · M12W2D6 · Final practical examination part 1: systems and network

Phase: Academic month · **Optional date:** 2027-09-12 · **Domain resources:** L01 / F08

1. Day / sequence	Global Day 321 of 365; calendar-free M12W2D6; optional date 2027-09-12.
2. Exact topic	Final practical examination part 1: systems and network
3. Exact subtopics	Unknown VM fault, packet capture, subnetting, network troubleshooting, Linux/Windows logs and scripting
4. Learning objective	Demonstrate practical competence without tutorial support under timed conditions
5. Primary lecture / reading + direct URL	Linux Journey `https://linuxjourney.com/` **Type:** Interactive text lessons · **Access:** Free
6. Exact section / lesson locator	Linux Journey and N06 Wireshark User's Guide as permitted references in open-book section. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Python 3 Official Tutorial `https://docs.python.org/3/tutorial/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 120 minutes: fix a network/OS fault, analyze a capture, write a short script and document evidence. **2.** Standard (~2 h): Complete 120 minutes: fix a network/OS fault, analyze a capture, write a short script and document evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 120 minutes: fix a network/OS fault, analyze a capture, write a short script and document evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-practical-1/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which evidence proves the fault was fixed rather than masked? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Practical score at least 35/50 and no safety violation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Linux Journey and N06 Wireshark User's Guide as permitted references in open-book section"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OverTheWire Bandit; write a five-line correction; retest next morning with a new 10-minute problem.

D322 · M12W2D7 · Final practical examination part 2: web and ethical assessment

Phase: Academic month · **Optional date:** 2027-09-13 · **Domain resources:** W06 / W17 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 322 of 365; calendar-free M12W2D7; optional date 2027-09-13.
2. Exact topic	Final practical examination part 2: web and ethical assessment
3. Exact subtopics	Authorization, scope, local app assessment, vulnerability validation, secure fix, report and retest
4. Learning objective	Demonstrate authorized testing and secure engineering competence
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP WSTG and E01 NIST SP 800-115 as permitted references. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 120 minutes on a local unseen app: scope, validate, remediate, retest and write a concise report — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 120 minutes on a local unseen app: scope, validate, remediate, retest and write a concise report — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Complete 120 minutes on a local unseen app: scope, validate, remediate, retest and write a concise report — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-practical-2/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which finding was rejected as a false positive and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Practical score at least 35/50 and scope boundaries are explicit Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP WSTG and E01 NIST SP 800-115 as permitted references"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 · Week 3 · daily cards

D323 · M12W3D1 · Final practical examination part 3: blue team and cloud

Phase: Academic month · **Optional date:** 2027-09-14 · **Domain resources:** B02 / CL03

1. Day / sequence	Global Day 323 of 365; calendar-free M12W3D1; optional date 2027-09-14.
2. Exact topic	Final practical examination part 3: blue team and cloud
3. Exact subtopics	Synthetic incident, KQL-style query, detection rule, ATT&CK mapping, cloud/IaC review and executive summary
4. Learning objective	Demonstrate detection, investigation and infrastructure-security competence
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL overview and B04 Sigma getting-started as permitted references. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete 120 minutes: investigate an incident, write/test one rule, review a cloud configuration and present a prioritized action plan. **2.** Standard (~2 h): Complete 120 minutes: investigate an incident, write/test one rule, review a cloud configuration and present a prioritized action plan. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete 120 minutes: investigate an incident, write/test one rule, review a cloud configuration and present a prioritized action plan. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-practical-3/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which data source is missing and how does it affect confidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Practical score at least 35/50 and all conclusions state uncertainty Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL overview and B04 Sigma getting-started as permitted references"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D324 · M12W3D2 · Final capstone defence and oral examination

Phase: Academic month · **Optional date:** 2027-09-15 · **Domain resources:** C15 / A08

1. Day / sequence	Global Day 324 of 365; calendar-free M12W3D2; optional date 2027-09-15.
2. Exact topic	Final capstone defence and oral examination
3. Exact subtopics	Architecture, threat model, original engineering, failures, validation, limitations and ethical boundaries
4. Learning objective	Defend the Month 11 capstone to a technical reviewer using evidence
5. Primary lecture / reading + direct URL	OWASP Application Security Verification Standard `https://owasp.org/www-project-application-security-verification-standard/` **Type:** Standard/reference · **Access:** Free
6. Exact section / lesson locator	OWASP ASVS Level 1 and C17 threat model as review checklist. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Deliver a 20-minute capstone defence and answer ten questions; provide links to tests, reports and clean setup evidence. **2.** Standard (~2 h): Deliver a 20-minute capstone defence and answer ten questions; provide links to tests, reports and clean setup evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Deliver a 20-minute capstone defence and answer ten questions; provide links to tests, reports and clean setup evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-defence.pdf and recording notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which claim in your report is weakest under questioning? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Score at least 75/100 and all high-risk claims have evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP ASVS Level 1 and C17 threat model as review checklist"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup OWASP Secure Coding Practices Quick Reference Guide; write a five-line correction; retest next morning with a new 10-minute problem.

D325 · M12W3D3 · Weekly final examination review and grade calculation

Phase: Academic month · **Optional date:** 2027-09-16 · **Domain resources:** C20 / F09

1. Day / sequence	Global Day 325 of 365; calendar-free M12W3D3; optional date 2027-09-16.
2. Exact topic	Weekly final examination review and grade calculation
3. Exact subtopics	Three practicals, capstone defence, monthly scores, portfolio quality and safety
4. Learning objective	Calculate the final 12-month mastery result using consistent weights
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework and the standard 100-point assessment rubric in Volume 6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete the final scorecard, classify competence as foundational/intermediate/advanced beginner specialization, and write a one-year reflection. **2.** Standard (~2 h): Complete the final scorecard, classify competence as foundational/intermediate/advanced beginner specialization, and write a one-year reflection. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete the final scorecard, classify competence as foundational/intermediate/advanced beginner specialization, and write a one-year reflection. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-final-scorecard.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which measurable competency improved most and which remains weakest? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All scores are evidence-linked and no competency is overstated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework and the standard 100-point assessment rubric in Volume 6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D326 · M12W3D4 · Final portfolio audit and sanitization

Phase: Academic month · **Optional date:** 2027-09-17 · **Domain resources:** W13 / F09

1. Day / sequence	Global Day 326 of 365; calendar-free M12W3D4; optional date 2027-09-17.
2. Exact topic	Final portfolio audit and sanitization
3. Exact subtopics	Secrets, personal data, IP addresses, logs, screenshots, authorization statements, reproducibility and licences
4. Learning objective	Ensure every public artifact is safe, professional and reproducible
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub code security documentation, secret scanning and repository security. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run a secret scan and manual checklist across all repositories; remove or redact sensitive data and verify every direct link. **2.** Standard (~2 h): Run a secret scan and manual checklist across all repositories; remove or redact sensitive data and verify every direct link. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run a secret scan and manual checklist across all repositories; remove or redact sensitive data and verify every direct link. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	portfolio-audit.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which file could still reveal real personal or lab-identifying information? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	No secrets or personal data remain and all links open Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub code security documentation, secret scanning and repository security"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D327 · M12W3D5 · CV, LinkedIn and technical story preparation

Phase: Academic month · **Optional date:** 2027-09-18 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 327 of 365; calendar-free M12W3D5; optional date 2027-09-18.
2. Exact topic	CV, LinkedIn and technical story preparation
3. Exact subtopics	One-page CV, project bullets, evidence links, technical skills, limitations and a 90-second personal narrative
4. Learning objective	Translate portfolio evidence into employer-readable language without exaggeration
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role vocabulary and official job-description analysis guidance from Volume 6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Draft a CV and LinkedIn summary; rewrite ten project bullets as problem-action-evidence-result statements. **2.** Standard (~2 h): Draft a CV and LinkedIn summary; rewrite ten project bullets as problem-action-evidence-result statements. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Draft a CV and LinkedIn summary; rewrite ten project bullets as problem-action-evidence-result statements. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	cv-draft.pdf and LinkedIn notes; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which sentence claims more than your evidence supports? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All claims map to a repository, report or assessment result Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role vocabulary and official job-description analysis guidance from Volume 6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D328 · M12W3D6 · Technical interview practice and scenario answers

Phase: Academic month · **Optional date:** 2027-09-19 · **Domain resources:** B15 / C18

1. Day / sequence	Global Day 328 of 365; calendar-free M12W3D6; optional date 2027-09-19.
2. Exact topic	Technical interview practice and scenario answers
3. Exact subtopics	OSI/TCP troubleshooting, Linux/Windows logs, SQLi/XSS prevention, detection logic, IAM, cloud shared responsibility and incident response
4. Learning objective	Practice explaining technical reasoning clearly under time pressure
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide and W17 OWASP Cheat Sheet Series for accurate terminology. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	MITRE ATT&CK `https://attack.mitre.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Record answers to ten scenario questions, self-score for structure, evidence, uncertainty and safety; repeat the three weakest. **2.** Standard (~2 h): Record answers to ten scenario questions, self-score for structure, evidence, uncertainty and safety; repeat the three weakest. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Record answers to ten scenario questions, self-score for structure, evidence, uncertainty and safety; repeat the three weakest. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	interview-practice.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which answer needs a concrete example rather than a definition? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least eight answers are clear, accurate and under three minutes Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide and W17 OWASP Cheat Sheet Series for accurate terminology"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D329 · M12W3D7 · Certification readiness and final decision

Phase: Academic month · **Optional date:** 2027-09-20 · **Domain resources:** A18 / C03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 329 of 365; calendar-free M12W3D7; optional date 2027-09-20.
2. Exact topic	Certification readiness and final decision
3. Exact subtopics	ISC2 CC, Security+ V7/V8, CCST Cybersecurity, SC-900, SC-200, eJPT, OSCP, AWS Security Specialty and BTL1
4. Learning objective	Decide if, when and whether to take a certification based on verified official facts and budget
5. Primary lecture / reading + direct URL	CompTIA Security+ V7 and V8 official pages https://www.comptia.org/en-us/certifications/security/ **Type:** Official certification page · **Access:** Paid exam; V7 US$404 from CompTIA blog dated January 2025; verify current store
6. Exact section / lesson locator	CompTIA Security+ V7/V8 official pages; A19 ISC2 CC; A17 Cisco CCST; A20 Microsoft SC-900. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISC2 Certified in Cybersecurity Exam Outline https://www.isc2.org/certifications/cc/cc-certification-exam-outline
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Re-check official costs/objectives, complete free practice assessments, estimate total cost including renewal, and write a go/no-go decision for the next six months — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Re-check official costs/objectives, complete free practice assessments, estimate total cost including renewal, and write a go/no-go decision for the next six months — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Re-check official costs/objectives, complete free practice assessments, estimate total cost including renewal, and write a go/no-go decision for the next six months — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	certification-final-decision.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What specific job or education requirement would justify paying for the exam now? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Decision cites current official sources and no exam is purchased merely for collection Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CompTIA Security+ V7/V8 official pages; A19 ISC2 CC; A17 Cisco CCST; A20 Microsoft SC-900"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Professor Messer SY0-701 Security+ Course Index; write a five-line correction; retest next morning with a new 10-minute problem.

Month 12 · Week 4 · daily cards

D330 · M12W4D1 · Year 2-5 long-term plan and 90-day sprint

Phase: Academic month · **Optional date:** 2027-09-21 · **Domain resources:** A08 / C20

1. Day / sequence	Global Day 330 of 365; calendar-free M12W4D1; optional date 2027-09-21.
2. Exact topic	Year 2-5 long-term plan and 90-day sprint
3. Exact subtopics	Specialization, peer group, research reading, open-source contribution, internship/job search, advanced labs and measurable milestones
4. Learning objective	Convert 12 months of foundations into a multi-year development plan
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering free book and A12 USENIX Security proceedings for research pathway examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a Year 2 plan, Year 3-5 milestones, a 90-day sprint, weekly metrics and a review date; define a meaningful peer group. **2.** Standard (~2 h): Write a Year 2 plan, Year 3-5 milestones, a 90-day sprint, weekly metrics and a review date; define a meaningful peer group. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a Year 2 plan, Year 3-5 milestones, a 90-day sprint, weekly metrics and a review date; define a meaningful peer group. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	year2-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which milestone measures technical impact rather than course completion? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is specific, realistic and does not promise top 1% status Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering free book and A12 USENIX Security proceedings for research pathway examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D331 · M12W4D2 · Monthly review, celebration and responsible handover

Phase: Academic month · **Optional date:** 2027-09-22 · **Domain resources:** C06 / F09

1. Day / sequence	Global Day 331 of 365; calendar-free M12W4D2; optional date 2027-09-22.
2. Exact topic	Monthly review, celebration and responsible handover
3. Exact subtopics	Achievements, evidence, failures, mental health, community contribution and next-year commitments
4. Learning objective	Recognize completion while planning sustainable long-term growth
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 and C20 NICE Framework for outcome categories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a final reflection, thank any mentors, publish a lessons-learned post with sanitized resources, and archive all private evidence. **2.** Standard (~2 h): Write a final reflection, thank any mentors, publish a lessons-learned post with sanitized resources, and archive all private evidence. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a final reflection, thank any mentors, publish a lessons-learned post with sanitized resources, and archive all private evidence. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	final-reflection.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would you change if you repeated the year, and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Reflection is honest and the public post leaks no private data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 and C20 NICE Framework for outcome categories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D332 · M12W4D3 · Master resource directory update and link re-verification

Phase: Academic month · **Optional date:** 2027-09-23 · **Domain resources:** C03 / A19

1. Day / sequence	Global Day 332 of 365; calendar-free M12W4D3; optional date 2027-09-23.
2. Exact topic	Master resource directory update and link re-verification
3. Exact subtopics	Important URLs, access type, course versions, verification date, backups and outdated material flags
4. Learning objective	Ensure the daily manual still points to current primary sources
5. Primary lecture / reading + direct URL	ISC2 Certified in Cybersecurity Exam Outline `https://www.isc2.org/certifications/cc/cc-certification-exam-outline` **Type:** Official exam outline · **Access:** Free
6. Exact section / lesson locator	ISC2 CC Exam Outline and A18 CompTIA Security+ V8 official page as examples of version-sensitive resources. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	ISC2 Certified in Cybersecurity official page `https://www.isc2.org/Certifications/CC`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Recheck all Tier 1 resource links, record HTTP status/date, replace broken links with verified backups, and update the verification register. **2.** Standard (~2 h): Recheck all Tier 1 resource links, record HTTP status/date, replace broken links with verified backups, and update the verification register. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Recheck all Tier 1 resource links, record HTTP status/date, replace broken links with verified backups, and update the verification register. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-resource-register.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which course or certification link changed version and could mislead future study? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Every mandatory resource is verified or has a documented backup and date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "ISC2 CC Exam Outline and A18 CompTIA Security+ V8 official page as examples of version-sensitive resources"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup ISC2 CC self-study resources; write a five-line correction; retest next morning with a new 10-minute problem.

D333 · M12W4D4 · Practice-exam strategy and question analysis

Phase: Academic month · **Optional date:** 2027-09-24 · **Domain resources:** C03 / C01

1. Day / sequence	Global Day 333 of 365; calendar-free M12W4D4; optional date 2027-09-24.
2. Exact topic	Practice-exam strategy and question analysis
3. Exact subtopics	Objective mapping, retrieval, distractor analysis, confidence marks, timeboxes, error taxonomy and retest intervals
4. Learning objective	Prepare for any future certification without reproducing proprietary exam questions
5. Primary lecture / reading + direct URL	ISC2 Certified in Cybersecurity Exam Outline `https://www.isc2.org/certifications/cc/cc-certification-exam-outline` **Type:** Official exam outline · **Access:** Free
6. Exact section / lesson locator	ISC2 CC official flashcards/quiz and A20 Microsoft SC-900 free practice assessment. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Professor Messer SY0-701 Security+ Course Index `https://www.professormesser.com/security-plus/sy0-701/sy0-701-video/sy0-701-comptia-security-plus-course/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a study plan for one chosen certification using only official free practice materials; take two 30-question self-generated quizzes and analyze errors. **2.** Standard (~2 h): Create a study plan for one chosen certification using only official free practice materials; take two 30-question self-generated quizzes and analyze errors. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a study plan for one chosen certification using only official free practice materials; take two 30-question self-generated quizzes and analyze errors. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-exam-strategy.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which error is due to wording rather than missing technical knowledge? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes at least three spaced retests and no brain-dump or pirated content Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "ISC2 CC official flashcards/quiz and A20 Microsoft SC-900 free practice assessment"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup ISC2 CC self-study resources; write a five-line correction; retest next morning with a new 10-minute problem.

D334 · M12W4D5 · Internship and entry-level application strategy

Phase: Academic month · **Optional date:** 2027-09-25 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 334 of 365; calendar-free M12W4D5; optional date 2027-09-25.
2. Exact topic	Internship and entry-level application strategy
3. Exact subtopics	Role targeting, application tracker, evidence links, cold outreach, referrals, interview prep and work-authorization checks
4. Learning objective	Build a realistic three-month application plan without guarantees
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role vocabulary and official career resources in Volume 6. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Identify 20 role types, draft a tracker, tailor one CV to a real job description, and list work-authorization/remote constraints for target regions. **2.** Standard (~2 h): Identify 20 role types, draft a tracker, tailor one CV to a real job description, and list work-authorization/remote constraints for target regions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Identify 20 role types, draft a tracker, tailor one CV to a real job description, and list work-authorization/remote constraints for target regions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	application-plan.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which role's requirements are closest to your portfolio evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is evidence-based and includes follow-up dates but no employment promise Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role vocabulary and official career resources in Volume 6"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D335 · M12W4D6 · Sustainable study system and burnout prevention

Phase: Academic month · **Optional date:** 2027-09-26 · **Domain resources:** C06 / F04

1. Day / sequence	Global Day 335 of 365; calendar-free M12W4D6; optional date 2027-09-26.
2. Exact topic	Sustainable study system and burnout prevention
3. Exact subtopics	Workload variants, recovery, backlog rules, spaced repetition, sleep, exercise, campus workload and exam seasons
4. Learning objective	Design a Year 2 schedule that can survive real life and university deadlines
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 Govern function for continuous improvement; C20 NICE for skill planning. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CS50's Introduction to Cybersecurity `https://cs50.harvard.edu/cybersecurity/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Define monthly/quarterly review rituals, a maximum backlog rule, minimum viable study session and two burnout warning signals with responses. **2.** Standard (~2 h): Define monthly/quarterly review rituals, a maximum backlog rule, minimum viable study session and two burnout warning signals with responses. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Define monthly/quarterly review rituals, a maximum backlog rule, minimum viable study session and two burnout warning signals with responses. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	sustainable-study-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What will you drop when university exams consume two weeks? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan includes explicit catch-up and reduced-intensity weeks Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 Govern function for continuous improvement; C20 NICE for skill planning"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D336 · M12W4D7 · Spaced revision and handover to final consolidation

Phase: Academic month · **Optional date:** 2027-09-27 · **Domain resources:** B03 / CL03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 336 of 365; calendar-free M12W4D7; optional date 2027-09-27.
2. Exact topic	Spaced revision and handover to final consolidation
3. Exact subtopics	All 11 modules, capstone, practical exams, portfolio and certification decision; light retrieval
4. Learning objective	Preserve the full year's knowledge and prepare for four weeks of independent consolidation
5. Primary lecture / reading + direct URL	SigmaHQ `https://sigmahq.io/` **Type:** Open-source detection rules · **Access:** Free/open-source
6. Exact section / lesson locator	SigmaHQ and B02 KQL overview for primary-specialization recall. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 20 mixed retrieval questions, create a final deck of 30 flashcards, clean snapshots and schedule the Week 49-52 challenges — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 20 mixed retrieval questions, create a final deck of 30 flashcards, clean snapshots and schedule the Week 49-52 challenges — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Do 20 mixed retrieval questions, create a final deck of 30 flashcards, clean snapshots and schedule the Week 49-52 challenges — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	M12-consolidation.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which three topics require review on days 7, 30 and 90? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Spaced repetition schedule exists and all lab resources are clean Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "SigmaHQ and B02 KQL overview for primary-specialization recall"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CISA Known Exploited Vulnerabilities Catalog; write a five-line correction; retest next morning with a new 10-minute problem.

Final consolidation — Weeks 49-52 and Day 365

This is the explicitly scheduled remainder of the 365-day plan: 12 academic months (336 days), four final consolidation weeks (28 days), and one graduation day. It is not vague "continue practising"; every day has a resource, practical task, evidence and assessment.

Week 49 · independent integration

D337 · W49D1 · Final consolidation kickoff: unseen integrated scenario

Phase: Final consolidation · **Optional date:** 2027-09-28 · **Domain resources:** C07 / B02

1. Day / sequence	Global Day 337 of 365; calendar-free W49D1; optional date 2027-09-28.
2. Exact topic	Final consolidation kickoff: unseen integrated scenario
3. Exact subtopics	Incident, network packet, Linux/Windows logs, cloud config, web finding and an ambiguous stakeholder request
4. Learning objective	Establish how you approach an unfamiliar problem before looking up answers
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 incident response lifecycle and C20 NICE work-role tasks. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Read the scenario once, list known/unknown facts, write five hypotheses and create an evidence plan before any tool use. **2.** Standard (~2 h): Read the scenario once, list known/unknown facts, write five hypotheses and create an evidence plan before any tool use. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Read the scenario once, list known/unknown facts, write five hypotheses and create an evidence plan before any tool use. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-case-brief.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which decision must be made before collecting data and why? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan is recorded before investigation and assumptions are labelled Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 incident response lifecycle and C20 NICE work-role tasks"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D338 · W49D2 · W49D2: systems and network investigation

Phase: Final consolidation · **Optional date:** 2027-09-29 · **Domain resources:** N06 / N07

1. Day / sequence	Global Day 338 of 365; calendar-free W49D2; optional date 2027-09-29.
2. Exact topic	W49D2: systems and network investigation
3. Exact subtopics	Host inventory, packets, subnetting, DNS, TLS, routing and service reachability
4. Learning objective	Resolve the systems/network portion of the integrated scenario
5. Primary lecture / reading + direct URL	Wireshark User's Guide `https://www.wireshark.org/docs/wsug_html_chunked/` **Type:** Official documentation · **Access:** Free
6. Exact section / lesson locator	Wireshark User's Guide and N09 Nmap Reference Guide for authorized lab scope. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Wireshark SampleCaptures `https://wiki.wireshark.org/SampleCaptures`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Work only with provided synthetic evidence; build a timeline, endpoint map and three alternative explanations. **2.** Standard (~2 h): Work only with provided synthetic evidence; build a timeline, endpoint map and three alternative explanations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Work only with provided synthetic evidence; build a timeline, endpoint map and three alternative explanations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-network.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which frame or log field changes your preferred explanation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Conclusions cite specific evidence and uncertainty Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Wireshark User's Guide and N09 Nmap Reference Guide for authorized lab scope"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Wireshark SampleCaptures; write a five-line correction; retest next morning with a new 10-minute problem.

D339 · W49D3 · W49D3: endpoint and identity investigation

Phase: Final consolidation · **Optional date:** 2027-09-30 · **Domain resources:** L09 / B05

1. Day / sequence	Global Day 339 of 365; calendar-free W49D3; optional date 2027-09-30.
2. Exact topic	W49D3: endpoint and identity investigation
3. Exact subtopics	Windows events, Sysmon, Linux auth logs, process trees, identity changes and privilege boundaries
4. Learning objective	Determine what happened on the endpoint without claiming more than telemetry shows
5. Primary lecture / reading + direct URL	Windows Event Log documentation `https://learn.microsoft.com/en-us/windows/win32/wes/windows-event-log` **Type:** Official reference · **Access:** Free
6. Exact section / lesson locator	Windows Event Log docs and L11 Sysmon documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SBousseaden EVTX-ATTACK-SAMPLES `https://github.com/sbousseaden/EVTX-ATTACK-SAMPLES`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a process/user timeline, identify privilege changes, and list the telemetry needed to close each gap. **2.** Standard (~2 h): Build a process/user timeline, identify privilege changes, and list the telemetry needed to close each gap. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a process/user timeline, identify privilege changes, and list the telemetry needed to close each gap. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-endpoint.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which event could be benign in one context and malicious in another? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Timeline is consistent and all claims have confidence labels Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Windows Event Log docs and L11 Sysmon documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Sysmon documentation and download; write a five-line correction; retest next morning with a new 10-minute problem.

D340 · W49D4 · W49D4: web and API evidence

Phase: Final consolidation · **Optional date:** 2027-10-01 · **Domain resources:** W06 / W01

1. Day / sequence	Global Day 340 of 365; calendar-free W49D4; optional date 2027-10-01.
2. Exact topic	W49D4: web and API evidence
3. Exact subtopics	HTTP requests, sessions, authorization, input validation, injection/XSS/SSRF indicators and server errors
4. Learning objective	Correlate application events with endpoint and identity evidence
5. Primary lecture / reading + direct URL	OWASP Web Security Testing Guide `https://owasp.org/www-project-web-security-testing-guide/` **Type:** Testing methodology · **Access:** Free
6. Exact section / lesson locator	OWASP Web Security Testing Guide and W17 OWASP Cheat Sheet Series. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	PortSwigger Web Security Academy `https://portswigger.net/web-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Analyze a provided synthetic request log, identify two suspicious patterns, propose safe validation tests and reject one false positive. **2.** Standard (~2 h): Analyze a provided synthetic request log, identify two suspicious patterns, propose safe validation tests and reject one false positive. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Analyze a provided synthetic request log, identify two suspicious patterns, propose safe validation tests and reject one false positive. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-web.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which request header or response status is decisive? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Findings are reproducible and one alternative explanation is retained Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Web Security Testing Guide and W17 OWASP Cheat Sheet Series"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D341 · W49D5 · W49D5: cloud, container and CI evidence

Phase: Final consolidation · **Optional date:** 2027-10-02 · **Domain resources:** CL03 / CL09

1. Day / sequence	Global Day 341 of 365; calendar-free W49D5; optional date 2027-10-02.
2. Exact topic	W49D5: cloud, container and CI evidence
3. Exact subtopics	IAM audit events, storage changes, Kubernetes RBAC, container logs and workflow artifacts
4. Learning objective	Determine whether the incident crossed into infrastructure or supply chain
5. Primary lecture / reading + direct URL	AWS Well-Architected Security Pillar `https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html` **Type:** Official architecture guide · **Access:** Free
6. Exact section / lesson locator	AWS Well-Architected Security Pillar and CL14 GitHub Actions hardening. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Kubernetes security documentation `https://kubernetes.io/docs/concepts/security/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review provided synthetic cloud/container evidence, map blast radius and identify one preventive and one detective control. **2.** Standard (~2 h): Review provided synthetic cloud/container evidence, map blast radius and identify one preventive and one detective control. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review provided synthetic cloud/container evidence, map blast radius and identify one preventive and one detective control. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-cloud.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which identity had the ability to change the build pipeline? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Scope is drawn with evidence and no live cloud resource is touched Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "AWS Well-Architected Security Pillar and CL14 GitHub Actions hardening"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup AWS security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D342 · W49D6 · W49D6: detection engineering and response

Phase: Final consolidation · **Optional date:** 2027-10-03 · **Domain resources:** B04 / B03

1. Day / sequence	Global Day 342 of 365; calendar-free W49D6; optional date 2027-10-03.
2. Exact topic	W49D6: detection engineering and response
3. Exact subtopics	Sigma/KQL rule, test data, true/false positives, containment recommendation and recovery validation
4. Learning objective	Turn the week's findings into an actionable detection and response
5. Primary lecture / reading + direct URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html` **Type:** Official rule reference · **Access:** Free
6. Exact section / lesson locator	Sigma getting-started and C07 NIST SP 800-61r3 response sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write one rule, ten labelled test events, an incident summary, a containment recommendation and a recovery test. **2.** Standard (~2 h): Write one rule, ten labelled test events, an incident summary, a containment recommendation and a recovery test. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write one rule, ten labelled test events, an incident summary, a containment recommendation and a recovery test. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-detection.yml and response.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which benign event could trigger the rule and how is it handled? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Rule passes tests and response preserves evidence Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Sigma getting-started and C07 NIST SP 800-61r3 response sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D343 · W49D7 · W49D7: weekly report, scoring and recovery

Phase: Final consolidation · **Optional date:** 2027-10-04 · **Domain resources:** B15 / C06 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 343 of 365; calendar-free W49D7; optional date 2027-10-04.
2. Exact topic	W49D7: weekly report, scoring and recovery
3. Exact subtopics	Integrated scenario, evidence quality, confidence, root cause, residual risk and lessons learned
4. Learning objective	Defend the week's conclusion and recover with lighter revision
5. Primary lecture / reading + direct URL	Microsoft SC-200 certification page `https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/` **Type:** Official certification page · **Access:** Free learning content; exam price region-based
6. Exact section / lesson locator	Microsoft SC-200 study guide investigation/reporting sections. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST Cybersecurity Framework 2.0 `https://www.nist.gov/cyberframework`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a five-page report, score it with the 100-point rubric, list three lessons, do 30 minutes of spaced recall and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a five-page report, score it with the 100-point rubric, list three lessons, do 30 minutes of spaced recall and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a five-page report, score it with the 100-point rubric, list three lessons, do 30 minutes of spaced recall and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W49-report.pdf and scorecard; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which conclusion would a skeptical reviewer challenge first? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Report passes 70/100 and includes a clear post-review action Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Microsoft SC-200 study guide investigation/reporting sections"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Microsoft Sentinel documentation; write a five-line correction; retest next morning with a new 10-minute problem.

Week 50 · independent integration

D344 · W50D1 · W50D1: research paper selection and reading plan

Phase: Final consolidation · **Optional date:** 2027-10-05 · **Domain resources:** A12 / A13

1. Day / sequence	Global Day 344 of 365; calendar-free W50D1; optional date 2027-10-05.
2. Exact topic	W50D1: research paper selection and reading plan
3. Exact subtopics	Primary source, research question, threat model, method, dataset, evaluation, limitations and reproducibility
4. Learning objective	Select one paper or engineering write-up relevant to the primary specialization
5. Primary lecture / reading + direct URL	USENIX Security conference papers `https://www.usenix.org/conferences/byname/164` **Type:** Conference proceedings · **Access:** Many papers freely available; conference attendance paid
6. Exact section / lesson locator	USENIX Security proceedings or Google Project Zero blog via A11; use one primary source only. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	arXiv cs.CR recent papers `https://arxiv.org/list/cs.CR/recent`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Choose a paper/write-up, record citation, prerequisites, three reading questions and a safe reproduction goal. **2.** Standard (~2 h): Choose a paper/write-up, record citation, prerequisites, three reading questions and a safe reproduction goal. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Choose a paper/write-up, record citation, prerequisites, three reading questions and a safe reproduction goal. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-reading-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What would count as successful reproduction without repeating dangerous behaviour? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Source is credible and reproduction is legal and isolated Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "USENIX Security proceedings or Google Project Zero blog via A11; use one primary source only"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Google Project Zero blog; write a five-line correction; retest next morning with a new 10-minute problem.

D345 · W50D2 · W50D2: primary-source close reading

Phase: Final consolidation · **Optional date:** 2027-10-06 · **Domain resources:** A08 / A11

1. Day / sequence	Global Day 345 of 365; calendar-free W50D2; optional date 2027-10-06.
2. Exact topic	W50D2: primary-source close reading
3. Exact subtopics	Terminology, claims, assumptions, experimental setup, baselines and statistical/technical evidence
4. Learning objective	Separate the authors' observations from their interpretations
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering book chapter on evaluating claims; A12 selected paper. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Google Project Zero blog `https://googleprojectzero.blogspot.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Annotate the paper's problem, method, data, results, limitations and five unfamiliar terms with official definitions. **2.** Standard (~2 h): Annotate the paper's problem, method, data, results, limitations and five unfamiliar terms with official definitions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Annotate the paper's problem, method, data, results, limitations and five unfamiliar terms with official definitions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-annotated-notes.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which claim depends on an assumption not tested by the authors? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Notes distinguish evidence, inference and speculation Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering book chapter on evaluating claims; A12 selected paper"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D346 · W50D3 · W50D3: reproduction design and lab safety

Phase: Final consolidation · **Optional date:** 2027-10-07 · **Domain resources:** A01 / B03

1. Day / sequence	Global Day 346 of 365; calendar-free W50D3; optional date 2027-10-07.
2. Exact topic	W50D3: reproduction design and lab safety
3. Exact subtopics	Environment, dependencies, synthetic data, metrics, controls, failure criteria and isolation
4. Learning objective	Design a minimal reproduction that tests one claim
5. Primary lecture / reading + direct URL	pwn.college `https://pwn.college/` **Type:** Hands-on university platform · **Access:** Free
6. Exact section / lesson locator	pwn.college dojo index for safe challenge infrastructure; A12 paper artifact if available. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a reproduction plan with exact inputs, expected outputs, abort conditions and a disposable VM/container boundary. **2.** Standard (~2 h): Write a reproduction plan with exact inputs, expected outputs, abort conditions and a disposable VM/container boundary. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a reproduction plan with exact inputs, expected outputs, abort conditions and a disposable VM/container boundary. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-reproduction-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which part of the original experiment cannot be reproduced safely or affordably? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan has a narrow claim, objective metrics and no unauthorized target Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pwn.college dojo index for safe challenge infrastructure; A12 paper artifact if available"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup pwn.college Dojo index; write a five-line correction; retest next morning with a new 10-minute problem.

D347 · W50D4 · W50D4: reproduction execution and evidence

Phase: Final consolidation · **Optional date:** 2027-10-08 · **Domain resources:** B02 / B04

1. Day / sequence	Global Day 347 of 365; calendar-free W50D4; optional date 2027-10-08.
2. Exact topic	W50D4: reproduction execution and evidence
3. Exact subtopics	Setup logs, commands, configuration, raw output, version numbers, errors and deviations
4. Learning objective	Reproduce one narrow result with complete evidence
5. Primary lecture / reading + direct URL	Kusto Query Language overview `https://learn.microsoft.com/en-us/kusto/query/?view=microsoft-fabric` **Type:** Official language docs · **Access:** Free
6. Exact section / lesson locator	KQL or primary-specialization official docs relevant to the paper's method. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Sigma rule creation guide `https://sigmahq.io/docs/guide/getting-started.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Run the minimal reproduction in an isolated lab, record all commands and outputs, and document every deviation from the paper. **2.** Standard (~2 h): Run the minimal reproduction in an isolated lab, record all commands and outputs, and document every deviation from the paper. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Run the minimal reproduction in an isolated lab, record all commands and outputs, and document every deviation from the paper. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-reproduction/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which deviation could explain a different result? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All evidence is captured and hashes/versions are recorded Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "KQL or primary-specialization official docs relevant to the paper's method"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup SigmaHQ; write a five-line correction; retest next morning with a new 10-minute problem.

D348 · W50D5 · W50D5: result comparison and negative findings

Phase: Final consolidation · **Optional date:** 2027-10-09 · **Domain resources:** A13 / A12

1. Day / sequence	Global Day 348 of 365; calendar-free W50D5; optional date 2027-10-09.
2. Exact topic	W50D5: result comparison and negative findings
3. Exact subtopics	Expected versus observed results, effect size, confidence, alternative explanations and limitations
4. Learning objective	Interpret reproduction results honestly even when they disagree with the paper
5. Primary lecture / reading + direct URL	arXiv cs.CR recent papers `https://arxiv.org/list/cs.CR/recent` **Type:** Preprint archive · **Access:** Free
6. Exact section / lesson locator	arXiv cs.CR for related work context only, not as authoritative evidence. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	USENIX Security conference papers `https://www.usenix.org/conferences/byname/164`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Compare results, calculate at least one simple metric, list three alternative explanations and decide what can be concluded. **2.** Standard (~2 h): Compare results, calculate at least one simple metric, list three alternative explanations and decide what can be concluded. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Compare results, calculate at least one simple metric, list three alternative explanations and decide what can be concluded. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-results.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which result is a negative finding and why is it still valuable? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Conclusion is supported by data and does not overclaim Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "arXiv cs.CR for related work context only, not as authoritative evidence"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup USENIX Security conference papers; write a five-line correction; retest next morning with a new 10-minute problem.

D349 · W50D6 · W50D6: open-source contribution or technical artifact

Phase: Final consolidation · **Optional date:** 2027-10-10 · **Domain resources:** F09 / B03

1. Day / sequence	Global Day 349 of 365; calendar-free W50D6; optional date 2027-10-10.
2. Exact topic	W50D6: open-source contribution or technical artifact
3. Exact subtopics	Issue selection, reproduction, small fix, documentation, tests, pull request or standalone utility
4. Learning objective	Contribute a small, useful, reviewable technical artifact
5. Primary lecture / reading + direct URL	Pro Git, 2nd Edition `https://git-scm.com/book/en/v2` **Type:** Free book / official reference · **Access:** Free
6. Exact section / lesson locator	Pro Git Chapters 2-3 and W13 GitHub security docs. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Choose one good-first issue or create a standalone detection/test utility; make a small change with tests and documentation. **2.** Standard (~2 h): Choose one good-first issue or create a standalone detection/test utility; make a small change with tests and documentation. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Choose one good-first issue or create a standalone detection/test utility; make a small change with tests and documentation. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-contribution/; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: What did maintainers or your own reviewer ask you to change? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Contribution is original, tested and free of secrets or personal data Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Pro Git Chapters 2-3 and W13 GitHub security docs"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Python for Everybody; write a five-line correction; retest next morning with a new 10-minute problem.

D350 · W50D7 · W50D7: research communication and recovery

Phase: Final consolidation · **Optional date:** 2027-10-11 · **Domain resources:** A08 / A11 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 350 of 365; calendar-free W50D7; optional date 2027-10-11.
2. Exact topic	W50D7: research communication and recovery
3. Exact subtopics	Abstract, problem, method, result, limitation, future work and safe disclosure of findings
4. Learning objective	Explain the week's research to a technical peer and recover
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering writing guidance and A12 paper structure examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Google Project Zero blog `https://googleprojectzero.blogspot.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a two-page technical note, record a five-minute explanation, schedule follow-up questions and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a two-page technical note, record a five-minute explanation, schedule follow-up questions and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write a two-page technical note, record a five-minute explanation, schedule follow-up questions and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W50-technical-note.pdf; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which sentence could mislead a reader about certainty? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Note is clear, modest and reproducible from the artifact Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering writing guidance and A12 paper structure examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Week 51 · independent integration

D351 · W51D1 · W51D1: teaching a foundational topic

Phase: Final consolidation · **Optional date:** 2027-10-12 · **Domain resources:** F01 / C20

1. Day / sequence	Global Day 351 of 365; calendar-free W51D1; optional date 2027-10-12.
2. Exact topic	W51D1: teaching a foundational topic
3. Exact subtopics	OS, networking, Linux/Windows, Python, core security, web, blue team or cloud concepts
4. Learning objective	Test mastery by teaching one topic without notes to a beginner
5. Primary lecture / reading + direct URL	CS50x 2026 OpenCourseWare `https://cs50.harvard.edu/x/` **Type:** University course · **Access:** Free; paid edX verified certificate optional
6. Exact section / lesson locator	CS50x notes or N01 Course Index for the selected topic. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a 20-minute lesson with one analogy, one diagram, one demo, three retrieval questions and common misconceptions. **2.** Standard (~2 h): Create a 20-minute lesson with one analogy, one diagram, one demo, three retrieval questions and common misconceptions. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a 20-minute lesson with one analogy, one diagram, one demo, three retrieval questions and common misconceptions. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-lesson.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which misconception did your explanation need to prevent? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Lesson is understandable and technically correct Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "CS50x notes or N01 Course Index for the selected topic"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CS50x Lectures 2026 (official YouTube playlist); write a five-line correction; retest next morning with a new 10-minute problem.

D352 · W51D2 · W51D2: code review and mentorship simulation

Phase: Final consolidation · **Optional date:** 2027-10-13 · **Domain resources:** P08 / W13

1. Day / sequence	Global Day 352 of 365; calendar-free W51D2; optional date 2027-10-13.
2. Exact topic	W51D2: code review and mentorship simulation
3. Exact subtopics	Diff reading, security impact, false positives, constructive feedback, tests and documentation
4. Learning objective	Review another student's sanitized code or your old Month 4 code
5. Primary lecture / reading + direct URL	OWASP Secure Coding Practices Quick Reference Guide `https://owasp.org/www-project-secure-coding-practices-quick-reference-guide/` **Type:** Checklist/reference · **Access:** Free
6. Exact section / lesson locator	OWASP Secure Coding Practices and W14 Semgrep documentation. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Perform a code review with five findings, one positive observation, suggested tests and respectful language. **2.** Standard (~2 h): Perform a code review with five findings, one positive observation, suggested tests and respectful language. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Perform a code review with five findings, one positive observation, suggested tests and respectful language. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-code-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which finding is a real risk and which is style preference? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Feedback is actionable, prioritized and technically justified Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "OWASP Secure Coding Practices and W14 Semgrep documentation"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup PortSwigger Web Security Academy; write a five-line correction; retest next morning with a new 10-minute problem.

D353 · W51D3 · W51D3: incident report peer review

Phase: Final consolidation · **Optional date:** 2027-10-14 · **Domain resources:** C07 / B17

1. Day / sequence	Global Day 353 of 365; calendar-free W51D3; optional date 2027-10-14.
2. Exact topic	W51D3: incident report peer review
3. Exact subtopics	Facts, assumptions, confidence, timeline, root cause, impact, recommendations and executive summary
4. Learning objective	Improve a synthetic incident report through peer-style critique
5. Primary lecture / reading + direct URL	NIST SP 800-61 Revision 3 `https://csrc.nist.gov/pubs/sp/800/61/r3/final` **Type:** Official guidance · **Access:** Free
6. Exact section / lesson locator	NIST SP 800-61r3 and B15 SC-200 study guide. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Review a intentionally flawed report, mark ten issues and rewrite the executive summary and recommendations. **2.** Standard (~2 h): Review a intentionally flawed report, mark ten issues and rewrite the executive summary and recommendations. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Review a intentionally flawed report, mark ten issues and rewrite the executive summary and recommendations. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-report-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which recommendation lacks evidence or an owner? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Revised report is concise and evidence-linked Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST SP 800-61r3 and B15 SC-200 study guide"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D354 · W51D4 · W51D4: interview scenario circuit

Phase: Final consolidation · **Optional date:** 2027-10-15 · **Domain resources:** C20 / W17

1. Day / sequence	Global Day 354 of 365; calendar-free W51D4; optional date 2027-10-15.
2. Exact topic	W51D4: interview scenario circuit
3. Exact subtopics	Technical, behavioural, troubleshooting, ethical and uncertainty questions
4. Learning objective	Practice communicating competence honestly under interview conditions
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks and B15 study guide for terminology. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	OWASP Cheat Sheet Series `https://cheatsheetseries.owasp.org/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Complete six timed interview scenarios; score answers for structure, evidence, safety, uncertainty and listening. **2.** Standard (~2 h): Complete six timed interview scenarios; score answers for structure, evidence, safety, uncertainty and listening. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Complete six timed interview scenarios; score answers for structure, evidence, safety, uncertainty and listening. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-interview-scores.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which answer needs a concrete lab example? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least five answers are clear and evidence-based Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks and B15 study guide for terminology"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D355 · W51D5 · W51D5: CV, portfolio and job-description mapping

Phase: Final consolidation · **Optional date:** 2027-10-16 · **Domain resources:** C20 / B15

1. Day / sequence	Global Day 355 of 365; calendar-free W51D5; optional date 2027-10-16.
2. Exact topic	W51D5: CV, portfolio and job-description mapping
3. Exact subtopics	Real job postings, required/must-have/nice-to-have skills, evidence mapping and gap plan
4. Learning objective	Analyze three current job descriptions without assuming they represent the whole market
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework https://www.nist.gov/itl/applied-cybersecurity/nice **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work roles; use live job postings saved with retrieval date in the workspace if available. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Microsoft SC-200 certification page https://learn.microsoft.com/en-us/credentials/certifications/security-operations-analyst/
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: For each of three postings, map ten requirements to portfolio evidence or a learning task; note visa/remote constraints separately. **2.** Standard (~2 h): For each of three postings, map ten requirements to portfolio evidence or a learning task; note visa/remote constraints separately. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): For each of three postings, map ten requirements to portfolio evidence or a learning task; note visa/remote constraints separately. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-job-analysis.csv; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which requirement is genuinely missing from your evidence? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Analysis distinguishes employer-specific wants from broad patterns and cites posting dates Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work roles; use live job postings saved with retrieval date in the workspace if available"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D356 · W51D6 · W51D6: professional network and community contribution

Phase: Final consolidation · **Optional date:** 2027-10-17 · **Domain resources:** C20 / B17

1. Day / sequence	Global Day 356 of 365; calendar-free W51D6; optional date 2027-10-17.
2. Exact topic	W51D6: professional network and community contribution
3. Exact subtopics	Local/international communities, CTFs, open-source, meetups, responsible communication and mentorship
4. Learning objective	Build a sustainable professional learning network without spam
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework community resources and E05 disclosure guidelines. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	CISA cybersecurity advisories `https://www.cisa.gov/news-events/cybersecurity-advisories`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Identify five credible communities or events, draft two useful questions or contributions, and plan one monthly participation habit. **2.** Standard (~2 h): Identify five credible communities or events, draft two useful questions or contributions, and plan one monthly participation habit. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Identify five credible communities or events, draft two useful questions or contributions, and plan one monthly participation habit. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-community-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which community gives feedback from people stronger than you? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan prioritizes feedback and contribution over self-promotion Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework community resources and E05 disclosure guidelines"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D357 · W51D7 · W51D7: weekly review, rest and Year 2 research questions

Phase: Final consolidation · **Optional date:** 2027-10-18 · **Domain resources:** A08 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 357 of 365; calendar-free W51D7; optional date 2027-10-18.
2. Exact topic	W51D7: weekly review, rest and Year 2 research questions
3. Exact subtopics	Teaching, review, interviews, job analysis and community; spaced repetition and recovery
4. Learning objective	Consolidate the week and generate long-term research questions
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering book and A12 proceedings for question inspiration. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write five research questions, review 20 flashcards, update the Year 2 plan, and rest or take a light lab walkthrough — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write five research questions, review 20 flashcards, update the Year 2 plan, and rest or take a light lab walkthrough — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Write five research questions, review 20 flashcards, update the Year 2 plan, and rest or take a light lab walkthrough — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W51-review.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: Which question is narrow enough for a 90-day investigation? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	At least two research questions are specific and measurable Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering book and A12 proceedings for question inspiration"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Week 52 · independent integration

D358 · W52D1 · W52D1: 90-day specialization sprint design

Phase: Final consolidation · **Optional date:** 2027-10-19 · **Domain resources:** C20 / A12

1. Day / sequence	Global Day 358 of 365; calendar-free W52D1; optional date 2027-10-19.
2. Exact topic	W52D1: 90-day specialization sprint design
3. Exact subtopics	One specialization, one secondary skill, problem, metrics, weekly milestones, artifacts and review cadence
4. Learning objective	Launch the first post-12-month sprint with realistic workload
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework and B03 SigmaHQ for primary specialization evidence. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	USENIX Security conference papers `https://www.usenix.org/conferences/byname/164`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a 90-day plan with weekly deliverables, leading/lagging metrics, reading list, feedback source and stop/continue criteria. **2.** Standard (~2 h): Write a 90-day plan with weekly deliverables, leading/lagging metrics, reading list, feedback source and stop/continue criteria. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a 90-day plan with weekly deliverables, leading/lagging metrics, reading list, feedback source and stop/continue criteria. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-90-day-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which metric measures independent technical impact? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Plan fits the chosen 1/2/3-4 hour track and has a review date Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework and B03 SigmaHQ for primary specialization evidence"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D359 · W52D2 · W52D2: advanced lab environment rebuild

Phase: Final consolidation · **Optional date:** 2027-10-20 · **Domain resources:** CL06 / W13

1. Day / sequence	Global Day 359 of 365; calendar-free W52D2; optional date 2027-10-20.
2. Exact topic	W52D2: advanced lab environment rebuild
3. Exact subtopics	Clean snapshots, reproducible setup, documentation, backups, resource budget and isolation
4. Learning objective	Build a stable environment for Year 2 independent work
5. Primary lecture / reading + direct URL	Docker documentation `https://docs.docker.com/` **Type:** Official documentation · **Access:** Free; Docker Desktop licensing may apply in some organizations
6. Exact section / lesson locator	Docker documentation and F12 VirtualBox User Manual snapshots/networking. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GitHub code security documentation `https://docs.github.com/en/code-security`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Rebuild or verify the lab from documentation, restore from clean snapshots, hash key artifacts and record costs/resources. **2.** Standard (~2 h): Rebuild or verify the lab from documentation, restore from clean snapshots, hash key artifacts and record costs/resources. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Rebuild or verify the lab from documentation, restore from clean snapshots, hash key artifacts and record costs/resources. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-lab-rebuild.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which step still requires undocumented manual work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Fresh setup succeeds and no active cloud bill remains Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Docker documentation and F12 VirtualBox User Manual snapshots/networking"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Docker Engine security documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D360 · W52D3 · W52D3: Year 2 reading and research pipeline

Phase: Final consolidation · **Optional date:** 2027-10-21 · **Domain resources:** A08 / A11

1. Day / sequence	Global Day 360 of 365; calendar-free W52D3; optional date 2027-10-21.
2. Exact topic	W52D3: Year 2 reading and research pipeline
3. Exact subtopics	Books, official docs, conference papers, advisories, incident reports, notes and hypothesis tracking
4. Learning objective	Create a repeatable reading and note-taking workflow
5. Primary lecture / reading + direct URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html` **Type:** Free book draft · **Access:** Free online draft; print edition paid
6. Exact section / lesson locator	Security Engineering, A12 USENIX Security, A13 arXiv cs.CR, B17 CISA advisories. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Google Project Zero blog `https://googleprojectzero.blogspot.com/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Build a reading queue with 12 sources, a note template, a hypothesis log and a monthly paper discussion goal. **2.** Standard (~2 h): Build a reading queue with 12 sources, a note template, a hypothesis log and a monthly paper discussion goal. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Build a reading queue with 12 sources, a note template, a hypothesis log and a monthly paper discussion goal. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-reading-pipeline.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which source is primary evidence and which is secondary commentary? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Queue is staged by prerequisite and every source has a purpose Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "Security Engineering, A12 USENIX Security, A13 arXiv cs.CR, B17 CISA advisories"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D361 · W52D4 · W52D4: advanced practice ladder

Phase: Final consolidation · **Optional date:** 2027-10-22 · **Domain resources:** A01 / A09

1. Day / sequence	Global Day 361 of 365; calendar-free W52D4; optional date 2027-10-22.
2. Exact topic	W52D4: advanced practice ladder
3. Exact subtopics	Guided reproduction, modified exercise, unfamiliar problem, experiment design, original tool/ research and peer defence
4. Learning objective	Map Year 2 practice from dependent to independent work
5. Primary lecture / reading + direct URL	pwn.college `https://pwn.college/` **Type:** Hands-on university platform · **Access:** Free
6. Exact section / lesson locator	pwn.college dojos and A03 OpenSecurityTraining2 for advanced progressions. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Nightmare binary exploitation course `https://guyinatuxedo.github.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create a six-level ladder with one concrete challenge per level for your chosen specialization. **2.** Standard (~2 h): Create a six-level ladder with one concrete challenge per level for your chosen specialization. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create a six-level ladder with one concrete challenge per level for your chosen specialization. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-practice-ladder.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which level marks the transition from learning to original contribution? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Ladder has objective evidence at each level Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "pwn.college dojos and A03 OpenSecurityTraining2 for advanced progressions"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup pwn.college Dojo index; write a five-line correction; retest next morning with a new 10-minute problem.

D362 · W52D5 · W52D5: technical impact and peer group definition

Phase: Final consolidation · **Optional date:** 2027-10-23 · **Domain resources:** C20 / A08

1. Day / sequence	Global Day 362 of 365; calendar-free W52D5; optional date 2027-10-23.
2. Exact topic	W52D5: technical impact and peer group definition
3. Exact subtopics	Specialization scope, peer group, impact metrics, reputation versus skill, review and feedback
4. Learning objective	Define what exceptional competence would mean in a realistic peer group
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework and A12 USENIX Security for peer-reviewed impact examples. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	Security Engineering, Third Edition (free online chapters) `https://www.cl.cam.ac.uk/~rja14/book.html`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Write a personal definition of impact, choose a peer group, list three measurable signals and three vanity metrics to ignore. **2.** Standard (~2 h): Write a personal definition of impact, choose a peer group, list three measurable signals and three vanity metrics to ignore. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Write a personal definition of impact, choose a peer group, list three measurable signals and three vanity metrics to ignore. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-impact-plan.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which signal would still matter if nobody publicly praised the work? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Definition separates technical impact from popularity Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework and A12 USENIX Security for peer-reviewed impact examples"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

D363 · W52D6 · W52D6: final archive, backup and privacy audit

Phase: Final consolidation · **Optional date:** 2027-10-24 · **Domain resources:** W13 / C24

1. Day / sequence	Global Day 363 of 365; calendar-free W52D6; optional date 2027-10-24.
2. Exact topic	W52D6: final archive, backup and privacy audit
3. Exact subtopics	All evidence, reports, repositories, snapshots, keys, credentials, personal data and retention
4. Learning objective	Preserve useful work while removing unnecessary sensitive data
5. Primary lecture / reading + direct URL	GitHub code security documentation `https://docs.github.com/en/code-security` **Type:** Official documentation · **Access:** Free features vary by repository visibility and plan
6. Exact section / lesson locator	GitHub security docs and F09 Pro Git. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	GDPR full text (EUR-Lex) `https://eur-lex.europa.eu/eli/reg/2016/679/oj`
8. Practical task (Core / Standard / Intensive)	**1.** Core (~1 h): complete one minimum instance of: Create an encrypted local archive, verify hashes, delete unnecessary secrets/data from old repositories, and write a retention schedule. **2.** Standard (~2 h): Create an encrypted local archive, verify hashes, delete unnecessary secrets/data from old repositories, and write a retention schedule. Then change one variable, document one error, and save all command output. **3.** Intensive (3-4 h): Standard (~2 h): Create an encrypted local archive, verify hashes, delete unnecessary secrets/data from old repositories, and write a retention schedule. Then change one variable, document one error, and save all command output. Rebuild from a blank file/VM, add one independent extension, consult one primary doc, and write a 150-word teach-back. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-archive-manifest.txt; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h · Standard: 2 h · Intensive: 3-4 h.
11. Revision / retrieval exercise	Closed-book: Which old artifact should be deleted rather than archived? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Restore test succeeds and no unencrypted secrets remain Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "GitHub security docs and F09 Pro Git"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup Semgrep documentation; write a five-line correction; retest next morning with a new 10-minute problem.

D364 · W52D7 · W52D7: one-year retrospective and public learning statement

Phase: Final consolidation · **Optional date:** 2027-10-25 · **Domain resources:** C06 / C20 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 364 of 365; calendar-free W52D7; optional date 2027-10-25.
2. Exact topic	W52D7: one-year retrospective and public learning statement
3. Exact subtopics	Achievements, evidence, failures, limitations, gratitude, next milestone and sustainable habits
4. Learning objective	Complete the 12-month curriculum with an honest retrospective
5. Primary lecture / reading + direct URL	NIST Cybersecurity Framework 2.0 https://www.nist.gov/cyberframework **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NIST CSF 2.0 outcome categories for broad reflection. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	NIST NICE Workforce Framework https://www.nist.gov/itl/applied-cybersecurity/nice
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Publish a sanitized lessons-learned post, update CV/portfolio links, record the next 90-day commitment and rest — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Publish a sanitized lessons-learned post, update CV/portfolio links, record the next 90-day commitment and rest — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Publish a sanitized lessons-learned post, update CV/portfolio links, record the next 90-day commitment and rest — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	W52-retrospective.md; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What is the most important behaviour to sustain, not just the most important topic learned? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	Post is honest, safe and includes at least three measurable Year 2 goals Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NIST CSF 2.0 outcome categories for broad reflection"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup CIS Controls Version 8; write a five-line correction; retest next morning with a new 10-minute problem.

D365 · D365 · D365: graduation checklist and next-day start

Phase: Graduation · **Optional date:** 2027-10-26 · **Domain resources:** C20 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 365 of 365; calendar-free D365; optional date 2027-10-26.
2. Exact topic	D365: graduation checklist and next-day start
3. Exact subtopics	All 365 daily assignments, 15 projects, four milestone gates, practical exams, capstone defence, portfolio, certification decision and Year 2 plan
4. Learning objective	Verify that every required artifact exists and decide the very first task of Year 2
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks and the master checklist in Volume 1. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	completion-certificate.md and final checklist; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What evidence best proves independent problem-solving today? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All mandatory items are present, files open, and tomorrow has one specific task Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks and the master checklist in Volume 1"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.

Day 365 · graduation and next-day start

D365 · D365 · D365: graduation checklist and next-day start

Phase: Graduation · **Optional date:** 2027-10-26 · **Domain resources:** C20 / B03 · LIGHT REVISION / CATCH-UP

1. Day / sequence	Global Day 365 of 365; calendar-free D365; optional date 2027-10-26.
2. Exact topic	D365: graduation checklist and next-day start
3. Exact subtopics	All 365 daily assignments, 15 projects, four milestone gates, practical exams, capstone defence, portfolio, certification decision and Year 2 plan
4. Learning objective	Verify that every required artifact exists and decide the very first task of Year 2
5. Primary lecture / reading + direct URL	NIST NICE Workforce Framework `https://www.nist.gov/itl/applied-cybersecurity/nice` **Type:** Framework · **Access:** Free
6. Exact section / lesson locator	NICE Framework work-role tasks and the master checklist in Volume 1. No video timestamp is asserted unless visible on the video page.
7. Supplementary resource + URL	SigmaHQ `https://sigmahq.io/`
8. Practical task (Core / Standard / Intensive)	**1.** Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. **2.** Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. **3.** Light Intensive (~2-3 h): Light Standard (~1.5-2 h): Light Core (~1 h): 20 min retrieval, 20 min minimal practical check — Run the final completion checklist, repair any missing artifact, present the master portfolio to the user, and schedule tomorrow's first 90-day sprint task — and 10 min evidence. Add one variation and one error-log correction. Add a blank-state repeat and a 100-word teach-back. Still a recovery day. **Safety:** use only synthetic, owned or explicitly authorized data; no public targets.
9. Expected output / evidence	completion-certificate.md and final checklist; plus a journal entry or Git commit with date and tool versions.
10. Estimated duration	Core: 1 h (light) · Standard: 1.5-2 h (light) · Intensive: 2-3 h (light).
11. Revision / retrieval exercise	Closed-book: What evidence best proves independent problem-solving today? Then explain your answer aloud in under two minutes and check against the primary source.
12. Completion criteria	All mandatory items are present, files open, and tomorrow has one specific task Evidence saved; safety pass; Core task complete.
13. If assessment fails / targeted revision	If below 70%: revisit the exact locator "NICE Framework work-role tasks and the master checklist in Volume 1"; use the supplementary resource as an alternate explanation; repeat the practical task with a new input; consult backup NIST Cybersecurity Framework 2.0; write a five-line correction; retest next morning with a new 10-minute problem.